AFRICA CENTRE OF EXCELLENCE FOR SUSTAINABLE POWER AND ENERGY DEVELOPMENT (ACE-SPED)

UNIVERSITY OF NIGERIA NSUKKA

ENVIRONMENTAL AND SOCIAL MANAGEMENT PLAN (ESMP) FOR CONSTRUCTION OF ACE-SPED BUILDING AND REHABILITATION OF HOSTELS

FINAL REPORT

JANUARY, 2023

LIST OF ACRONYMS AND ABBREVIATIONS

AAU	Association of African Universities
ACE	Africa Center of Excellence
ACE-SPED	Africa Center of Excellence for Sustainable Power and Energy Development
ECOWAS	Economic Community of West African States
EA	Environmental Assessment
EIA	Environmental Impact Assessment
ENSEPA	Enugu state environmental protection agency
ESFS	Environmental and Social Data Sheet
ESIS	Environmental and Social Information Sheet (ESIS)
ESMF	Environmental and Social Management Framework
ESMEnv	Enugu state ministry of environment
ESMP	Environmental and Social Management Plan
ESO	Environmental safeguard officer
ESWAMA	Enugu State Waste Management Agency
FMEnv	Federal Ministry of Environment
GHG	Greenhouse gas
GBV/SEA	Gender based violence and sexual exploitation and assault
GRC	Grievance redress committee
GRM	Grievance redress mechanism
IDA	International Development Assistance
MDAs	Ministries, Agencies and Departments
MESTI	Ministry of Environment, Science, Technology and Innovation
NCE	National Council on Environment
NEMA	National Emergency Management Agency
NESREA	National Environmental Standards and Regulations Enforcement Agency
NPE	National Policy on the Environment
NUC	Nigerian Universities Commission
OP	Operational policy
RFU	Regional Facilitation Unit
SSF	Simplified Screening Form
SSO	Social safeguard officer
ToR	Terms of Reference
UNN	University of Nigeria Nsukka
WB	World Bank
WHO	World Health Organization

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EXECUTIVE SUMMARY

ES 1: Project Background

The World Bank is assisting in establishing the Africa Center of Excellence for Sustainable Power and Energy Development (ACE-SPED) which was conceptualized to proffer sustainable solutions to some development challenges peculiar to the Sub-Saharan Africa which include lack of access to reliable electric power supply, poor diffusion of renewable and alternative energy technologies, as well as shortage of high-level skilled manpower to deploy and manage the power supply system. The University of Nigeria Nsukka (UNN) is among the 17 higher institutions selected to benefit from the project.

Project components

ACE-SPED has three (3) components namely; component1: Establishing new Africa Centers of Excellence and scaling up well-performing existing Africa Centers of Excellence (ACEs) for development impact, Component 2: Regional partnership and scholarship 3: This component will support regional policy making for higher education and regional project monitoring and facilitation.

Rationale For Environmental and Social Management Plan (ESMP)

The activities associated with the project activities will triggers World Bank safeguards Operational Policy (OP) OP/BP 4.01 (Environmental Assessment), that covers impacts on the environment, human health and safety of the project. The project is classified under the World Bank safeguards policy as category B, as its adverse effects on the population or areas of environmental importance are limited, site-specific, and likely reversible, and the mitigation measures can be easily designed and implemented. The ESMP is the appropriate instrument to address the policies triggered by the project.

Objective of the ESMP

The objective of the ESMP is to ensure that all adverse Environmental & Social impacts likely to arise from the project activities are identified and addressed using appropriate mitigation measures integrated into the project to ensure protection of the environment and the people.

ES.2: Policy, Legal and Institutional Framework

The ESMP complies with the provisions of the World Bank OP 4.01, Nigerian Environmental Impact Assessment Act CAP. E12 L.F.N. 2004, the National Policy on the Environment, Harmful Waste (Special Criminal Provisions) ACT CAP H1, LFN 2004, National Air Quality Act, 1991; Employee's Compensation Act 2010; Child Right Act 2003; Labour Act - CAP. L1 L.F.N. 2004, and Violence Against Persons (prohibition) 2015. In terms of institutional framework, the coordination and operational role of this ESMP lies on the PIU within the UNN ACE-SPED. Other institutions providing regulatory supervision and oversight include the Federal Ministry of Environment, Enugu state ministry of Environment, NESREA, ESWAMA and the World Bank

ES 3: Environmental & Social baseline status of the project Area
Survey of the environmental and social conditions of the project area were carried out. The results are presented in sections 3.2 and 3.3, which serve as a baseline for measuring variations in environmental and social condition which may be attributed to the impact of the ACE-SPED project activities. Most social and environmental parameters indicate a no harm environmental and social conditions.

ES 4: Impact Assessment & Results: This is concerned with Methods and Techniques Used in identifying and Assessing the Environmental and Social Impacts of the project. Impact identification was achieved through public consultation, reconnaissance survey and use of checklist/survey questionnaire. Assessment of impact was carried out using Risk Assessment Matrix Tool. Major positive and adverse impacts of the project are described below:

Positive Impacts	
- The project will enhance impactful educational, research, development and training activities in Electric power systems development; Renewable energy, waste-to-energy and energy conservation; Energy resources assessment and forecasting; Sustainable energy materials and Energy Policy, Regulation and Management. - It will contribute to the overall socioeconomic well-being of the benefiting host community through the provision of employment opportunities. - During construction, some locals will be engaged as semi-skilled workers (e.g. masons, carpenters, electrical technicians, welders, etc.) - There will also be indirect opportunities for local contractors, businesses and food sellers amongst others. - Improved research staff capacity to engage in quality and novel research	
Adverse Impacts	
Environmental Impacts	**Social Impacts**
- Increase in fugitive dust emission during pre-construction and construction phases of the rehabilitation. - Waste generation including vegetation waste, construction wastes, municipal wastes and e-wastes - Possibility of seepage of fuel from machineries and effluent discharge into the environment, thus impacting on the environmental quality. - Loss of species diversity and abundance, including soil organisms, fungi, invertebrates, and bacteria due to site clearing	- Noise and vibration from heavy-duty project vehicles and equipment resulting in nuisance to residents and students. - Loss of employment for labourers after the completion of the construction and rehabilitation works. - Disruption of traffic movement. - Occupational Health and Safety risks. - Spread of sexually transmitted diseases (HIV, HB, Gonorrhea, etc.) - Increase in insecurity as a result of influx of migrant workers

- Air quality deterioration	- Gender-based violence and sexual molestation may result from interaction of construction staff, migrant workers and female students

ES 5: Mitigation Measures: Mitigation measures are contained in the body of this report including the following:

Impacts	Mitigation Measures
GBV/SEA	✓ Develop Codes of Conduct for civil works contractors with prohibitions against GBV/SEA, ✓ Strengthen/train the safeguards unit to handle issues on GBV/SEA and to manage social risks associated with GBV/SEA in the project. ✓ Address risks of GBV/SEA by providing guidance, awareness, and dissemination of relevant GBV/SEA materials to the people living in the project area including students
Waste Management	✓ Implement the Waste Management Plan
Air Quality	✓ Sprinkle water via spraying devices to limit dusts.
OHS	✓ Ensure the provision and use of PPEs
Soil contamination	✓ Limit zone of vehicle and equipment weight impacts (designate an area for parking and stacking equipment) Service vehicles; ✓ undergo vehicle emission testing (VET) and vehicle exhaust screening
Loss of employment by local workers	✓ Provide certificates/Testimonials to local and trained workers to help them secure employment

ES 6: The ESMP and Cost; This section proffered specific mitigation measures to each of the identified adverse impacts, assigned cost of mitigation and responsibilities for both mitigation and monitoring. The cost of implementing the ESMP is described as follows:

The total cost for the ESMP implementation and monitoring is **N13,936,000,** out of which N7,230,000 will be embedded in the contractors contracts, while the remaining **N6,706,000.00 (US$ 14,902)** is the component to be set aside for PIU ESMP implementation and monitoring**.).** The breakdown of the estimated budget is presented in the Table below:

Item	Responsibility	Cost Estimate in Nigerian Naira (N)	Cost Estimate in US Dollars (US$)*
Mitigation: ACE SPED Building construction	Contractor	Part of construction contract cost (N4,510,000)	
Mitigation: Hostels Renovation	Contractor	Part of renovation contract cost (N2,720,000)	
Monitoring (ACE-SPED Building and Hostel Renovation)	PIU	2,760,000	6,133
Capacity Building	SPIU, ENSMEnv and other relevant MDAs	1,500,000	3,333
GRM (10% of cost of total mitigation	PIU	**723,000**	**1,606**

GBV/SEA (10% of cost of total mitigation	PIU	**723,000**	**1,606**
Disclosure	PIU/FMEnv	1,000,000	2,222
Total		**6,706,000**	**14,902**

*1 US$ =N450

ES 7: ESMP Disclosures Plan

s/n	Action	Responsibility
1	Disclosure on 2 national newspapers	PIU/FMEnv
2	Disclosure at the Enugu State Ministry of Environment	PIU
3	Disclosure at Nsukka LGA office	ESO/PIU
4	Disclosure at Deputy VCs office, UNN	ESO/PIU
5	Disclosure at the World Bank external website	TTL/WB

ES7: Implementation Schedule

S/N	Activity Description	Responsible	Preconstrution (Month)					Construction (Month)									Operation		
			1	2	3	4	5	1	2	3	4	5	6	7	8	9			
1.	Clearance and Formal Disclosure of ESMP	PIU																	
2.	Inclusion of Environmental & Social Requirements in Bid Docs	PIU																	
3.	Allocating Budget for ESMP	PIU																	
4.	Appointing Support Staff for ESMP	PIU																	
5.	Review & Approval of Contractor's ESMP, Waste & Safety Plan	PIU																	
6.	Finalization of Engineering Designs	PIU/Engineering Design Consultant																	
7.	Environmental and Social Training	E&S Consultant																	
8.	Mobilization to site	Contractor																	
9.	Site Clearing	Contractor																	
10.	Construction Phase	Contractor																	
11.	Implementation of Mitigation	Contractor																	
12.	Supervising ESMP Implementation	PIU																	
13.	Monitoring & Reporting on ESMP Implementation	PIU/Relevant MDAs																	
14.	Environmental and Social Auditing	PIU/E&S Consultant																	

ES8: Grievance Redress Mechanism

This section provides procedures and 2 levels/channels for addressing disputes/grievances. They are the project level and the Senate channel. It recommends that an aggrieved person should channel his/her complaint first to the project level grievance redress committee (GRC) for hearing and redress; where he/she is dissatisfied with the outcome of the intervention of the committee, he/she can take the matter further to the Senate GRC. When he/she is not satisfied also with the outcome of the resolution at this level, the person has the right to seek for redress in the court of

law. The chapter provides for how matters can be received and handled including time period allowed to conclude all matters.

ES 9: Consultation with Stakeholders: This ESMP was prepared following identification and consultation with stakeholders. The consultation entailed in-depth interview with representatives of key stakeholders and a general consultation workshop which was convened at the Faculty of Engineering building. Major concerns identified included traffic and pedestrian obstruction; particulate dust during construction as well as stones flying from big truck when trucks begin to convey granite to the project site. Other concerns were risks of insecurity and Gender base violence and sexual exploitation as a result of presence of contractor workers and vendors. All concerns were addressed and factored into the design of the ESMP.

CHAPTER 1: INTRODUCTION

1.1Background

The Africa Center of Excellence for Sustainable Power and Energy Development (ACE-SPED) is based in the main campus of the University of Nigeria, Nsukka. The Centre was conceptualized to proffer sustainable solutions to some development challenges peculiar to the Sub-Saharan Africa region and it draws her staff from the University of Nigeria and operates based on already established managerial and fiduciary practices in the University. These developmental challenges faced by majority of countries in the region include lack of access to reliable electric power supply, poor diffusion of renewable and alternative energy technologies, as well as shortage of high-level skilled manpower to deploy and manage the power supply system.
The fundamental aims of ACE-SPED are; to carry out impactful educational, research, development and training activities in five major thematic research areas, namely (i) Electric power systems development; (ii) Renewable energy, waste-to-energy and energy conservation; (iii) Energy resources assessment and forecasting; (iv) Sustainable energy materials and (v) Energy Policy, Regulation and Management. Using a student-centered, interactive and engaging educational curriculum cum robust Research & Development themes with strong practice orientation, ACE-SPED plans to equip its Master's and PhD degree graduates to deliver great impact on sustainable power and energy development in the Sub-Saharan Africa region in the medium to long terms. These justify the urgency for an enabling infrastructure for administration, research, teaching, learning and hostel accommodation.

The ACE-SPED UNN building project involves the construction of the main building and the upgrade and rehabilitation of the existing hostels to be utilized for ACE-SPED UNN activities.

1.2 Objective and Rationale for the ESMP

The activities associated with the project activities such as the construction of laboratories, workshops, theatre and conference works block and the rehabilitation of existing school hostels may result in varying degrees of adverse impacts on the bio-physical, social environment, health and safety, and thus, triggers World Bank safeguards Operational Policy (OP) OP 4.01 (Environmental Assessment) and OP 4.11 (Physical and Cultural Resources),that covers impacts on the environment, human health and safety of the project. The project is classified under the World Bank safeguards policy as category B, as its adverse effects on the population or areas of environmental importance are limited, site-specific, and likely reversible, and the mitigation measures can be easily designed and implemented. The ESMP is the appropriate instrument to address the anticipated impacts associated with the project considering that the precise location of the site is now known.
The broad objective of the ESMP includes:

- Potentially screen out environmentally unsound activities
- Identify and Predicts significant adverse impacts
- Identifies mitigation measures to reduce, offset, or eliminate adverse impacts
- Propose mitigation measures to the identified impacts

- Engages and informs potentially affected communities and individuals on the impacts of the project
- Present the costs of implementing the proposed mitigation measures;

1.3 Scope of the ESMP

The core tasks for the ESMPs shall include:

- Assess the potential environmental and social impacts related to the construction and rehabilitation activities and recommend adequate mitigation measures, including costs estimation.
- Identify responsibilities and actors for the implementation of proposed mitigation measures.
- Assess the resources and competencies available to implement the proposed mitigation measures, and suggest recommendation in terms of training and capacity building, and estimate their costs.
- Develop an Environmental and Social Management Plan (ESMP) for the work.

1.4 Description of Methodology and Approach

The methodology of this study involves methods for impact identification and the method for assessment of potential impacts of the project. Impacts identification was achieved through public consultation, reconnaissance survey, sampling and measurement of air pollutants of the site, noise level, soil and water quality and use of checklist/questionnaire to identify socioeconomic issues that may affect perceptions and conditions of health and living conditions of the students, residents and beneficiaries of the ACE-SPED. Impact assessment was carried out using Risk Assessment Matrix tool which involves probability and consequences of occurrence of risk and impacts. These tools are contained in this report as appendices 2A and 2B respectively.

1.5 Discussion of the World Bank safeguard policies triggered by the Project and the proposed sub-project activity

The World Bank has 10+2 Environmental and Social safeguard policies, designed to help prevent and mitigate undue harm to people and their environment in the development process and ultimately ensuring that environmental and social issues are addressed throughout the project life cycle of a World Bank – financed project. Specifically, the proposed project has triggered the following policies: OP/BP 4.01: Environmental Assessment; OP/BP 4.11: Physical and Cultural Resources; 2010 Access to Information Policy.

World Bank Group Environmental, Health, and Safety Guidelines

World Bank Group Environmental, Health and Safety (EHS) Guidelines are technical reference documents with general and industry-specific examples of good international industry practice. The EHS guidelines contain the performance levels and measures that are normally acceptable to the World Bank Group and that are generally considered to be achievable for infrastructural developments

1.6 Institutional Framework:

Relevant to the implementation and hand on oversight of this project are key MDAs at Federal and state levels. These MDAs and their institutional roles are discussed below:

1.6.1 Federal Ministry of Environment (FMEnv)

The Federal Ministry of Environment (FMEnv) which was formerly known as the Federal Environmental Protection Agency (FEPA) was established in 1999 through Decree No. 58 of 1988 as amended by Decree No. 59 of 1992. The Ministry is the statutory government institution mandated to coordinate environmental protection and natural resources conservation for sustainable development in Nigeria. Some of the other mandates of the Ministry include:

- Advising the Federal Government on national environmental policies and priorities, conservation of natural resources, sustainable development as well as scientific and technological activities affecting the environment and natural resources; and
- Prescribing standards and formulating regulations on water quality, effluent limitations, air quality, atmospheric protection, ozone protection, noise control as well as the removal and control of hazardous substances.

As established in the EIA Act 86 of 1992, all public and private projects likely to negatively affect the environment must be submitted to the EIA process administered by the FMEnv including infrastructural development and agricultural activities. The EIA Division of the Environmental Assessment (EA) Department at the FMEnv is the main administrative unit responsible for overseeing the EIA process, preparing documents and developing the procedures necessary for decision making on project certification by the Minister of Environment.

1.6.2 National Environmental Standards and Regulations Enforcement Agency (NESREA)

The National Environmental Standards and Regulations Enforcement Agency (NESREA) was established by the NESREA Act of 30th July 2007 as an Agency of the FMEnv. NESREA is charged with the responsibility of enforcing all environmental laws, guidelines, policies, standards and regulations in Nigeria. It also has the responsibility to enforce compliance with provisions of international agreements, protocols, conventions and treaties on the environment to which Nigeria is a party.

1.6.3 Enugu State MDAs Relevant to the Project

In Enugu state, the Ministry of Environment and Mineral Resources has the responsibility for the general protection and development of the environment, conservation of biodiversity, natural resources and sustainable development. Other State MDAs relevant to the project include Enugu State Ministry of Agriculture and Rural Development (ENMARD), Ministry of Works and Infrastructure, Ministry of Lands and Urban Development, Enugu State Waste Management Authority (ESWAMA) and State Emergence Management Agency. The Enugu State Ministry of Environment and Mineral Resources established by an Act of Parliament in 2004 has the following functions:

i. Liaising with the Federal Ministry of Environment, (FMEnv) to achieve a healthy and better management of the environment via development of National Policies on Environment;

ii. Liaising with FMEnv and other National Directorates/Agencies in the performance of environmental functions including environmental education/awareness to the citizenry;

iii. Responsibility for monitoring waste management standards; iv. Responsibility for general environmental matters in the State, and;

iv. Monitoring the implementation of EIA/ESMP studies and other environmental studies for all development projects in the State;
v. Management of open space; vii. Setting standards for mitigation of climate change;
vi. Setting standards and guidelines for fumigation of premises; ix. Urban sanitation and beautification; x. Urban sewage management and control;
vii. Regulation of outdoor advertising in the state.

1.7 Legal and Regulatory Frameworks

The legal and regulatory framework guiding the preparation and implementation of this ESMP are the Nigerian laws, the Enugu State Government policies, World Bank Safeguard policies and other relevant international guidelines applicable to the sub-project intervention. The project will comply with the Nigerian Environmental Impact Assessment Act CAP. E12 L.F.N. 2004 and other relevant laws and policies as stipulated below:

Regulatory Framework	**Description**
Policies	
National Policy on the Environment, 1989. Revised 2016	The policy identifies key sectors requiring integration of environmental concerns and sustainability with development and presents their specific guidelines
National Policy on Occupational Safety and Health, revised 2020	This policy was approved by the Federal Executive Council (FEC) in September 2020. It provides a guide for voluntary compliance and serves as a basis for occupational health and safety programs for workers even under such development projects
National Gender Policy (2006)	Provides a framework for ensuring gender inclusion and sensitivity in developmental plans and programs at the national and sub-national levels.
National Inclusive Education Policy (2001)	This national policy provides that education must be inclusive for all children including those with disabilities. Children with disabilities have the right to qualitative, functional and effective basic education. The policy ensures that systems and strategies are modified to provide a barrier free environment for all learners with disabilities.
Sexual Harassment (SH) Policy	The policy is dedicated to enlightening its public on the evils of sexual harassment and other associated anti - social conduct Sexual harassment has very grave effects on the overall productivity of staff and students in the university
Acts	
Environmental Impact Assessment (EIA) Act CAP E12 LFN 2004	The Environmental Impact Assessment (EIA) Act CAP E12 LFN 2004 provides guidelines for activities of development projects for which EIA is mandatory in Nigeria. According to the act, category II projects such as the ACEPUTOIR requires only a partial EIA/EMP, which will focus on mitigation and Environmental planning measures,
Workers Compensation Act (2010)	The Workmen's Compensation Act makes provisions for the payment of compensation to workmen for injuries suffered in the course of their employment
The Violence Against Persons Prohibition Act (VAPP) (2015)	This act prohibits all forms of violence against private and public life and provides maximum protection and effective remedies for victims and punishment of offenders. Nigeria's national government has taken steps to penalize and address GBV and SEA.
Regulations	
National Environmental (Sanitation and	The purpose of the Regulation is the adoption of sustainable and environment friendly practices in environmental sanitation and waste management to minimize pollution. The

Regulatory Framework	Description
Wastes Control) Regulations (2009)	Instrument amongst others makes provisions for the control of solid wastes and hazardous wastes.
National Environmental (Soil Erosion & Flood Control) Regulations (S.I. 12) 2011	The purpose of these Regulations is to establish technically feasible and economically reasonable standards and procedures to achieve appropriate level of management and conservation practices to abate soil erosion, siltation and sedimentation of the waters of Nigeria, due to soil erosion and flood aggravated by non-agricultural earth-disturbing activities.
National Environmental Standards and Regulations Enforcement Agency (NESREA)	The National Environmental Standards and Regulations Enforcement Agency (NESREA) was established by the NESREA Act of 30th July 2007 as an Agency of the FMEnv. NESREA is charged with the responsibility of enforcing all environmental laws, guidelines, policies, standards and regulations in Nigeria. It also has the responsibility to enforce compliance with provisions of international agreements, protocols, conventions and treaties on the environment to which Nigeria is a party.

1.8 Structure of ESMP Report

The ESMP Report is structured into eight chapters as follows:

Chapter One: provides a background to the project and discusses the project components, rationale for the ESMP, objectives, approach and methodology and the structure of the ESMP Report.

Chapter Two: is the project description which provides descriptions of the sub-project activities

Chapter Three: Baseline description of the project area, including environment and social conditions.

Chapter Four: deals with assessment of potential environmental and social impacts and methods and techniques used in assessment.

Chapter Five: discusses the ESMP, presents the monitoring plan and outlines the roles, responsibilities and cost of implementing the management actions and monitoring plan.

Chapter Six: presents grievance redress mechanism

Chapter Seven: discusses the public consultation process and highlights the concerns, and how those concerns were addressed and,

Chapter Eight presents the conclusion and recommendations.

CHAPTER 2: PROJECT DESCRIPTION

2.1 Description of the UNN ACE-SPED Building

The situational analysis of the ACE-SPED UNN building project is presented in this chapter. The UNN ACE-SPED project is sited on a 25,640.77sqm land located at 6.86ºN latitude and 7.39ºE longitude. The design layout for the building is shown in Figure 1 below.

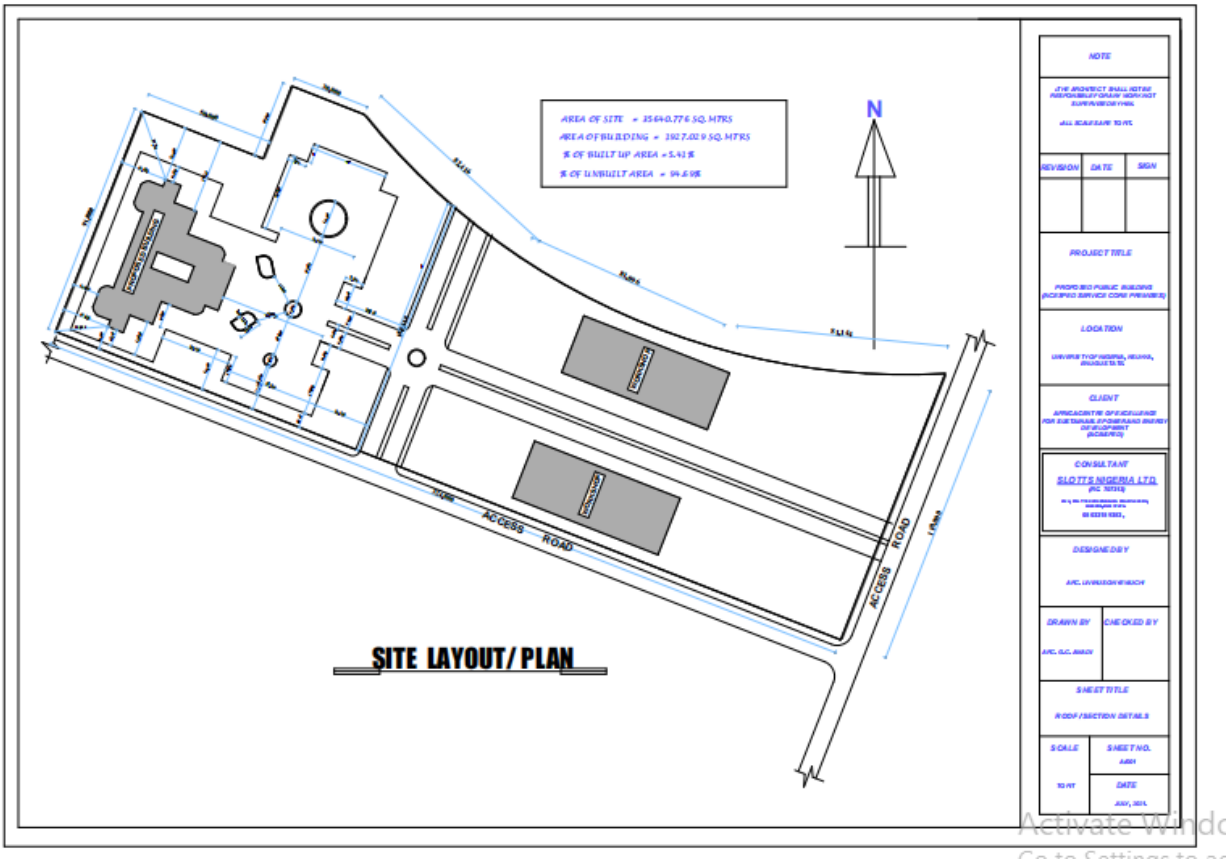


Figure 1: Site Layout design of the ACE-SPED Building
Source: UNN ACE-SPED ESMP Team (2022)

2.1.1 Components of the UNN ACE-SPED Building Construction

Under the ACE-SPED civil works construction, the University of Nigeria Nsukka intends to undertake the following:

- Construction of Director's office, administrative and Professorial office blocks Construction of engineering laboratories
- Construction of lecture theatre and conference room
- Construction of postgraduate students' library
- Construction of workshops

2.1.2 Activities of the Construction Works

- The activities of the project will involve land clearing of about 2600sqm for project buildings and corridor for movement of vehicles and equipment.
- The land clearing may be associated with removal of vegetal cover and shrubs which endangers the flora and fauna species in the area
- There is however, no surface water, market, private asset or other ecological sensitive resource to be impacted.
- Settlement around the site is the UNN staff quarter at Eni-Njoku Street which accommodates 8 residential buildings and about 40 household members in all.

Figure 2: Vegetation for Clearing at Proposed Project Site

The building will house the Director's office, administrative and Professorial office, engineering laboratories, lecture theatre and conference room as well as postgraduate students' library. The workshop building will be a stand-alone block also located within the layout.
The Activities to be undertaken during the implementation of the proposed ACE-SPED project are broadly classified into three (3) phases, namely; pre-construction, construction and operation phases. Activities in each phase are described in the following sub-sections.

Pre-construction Phase

This project entails the construction of new buildings as listed above and as such it is envisaged not to require land acquisition. This stemmed from the fact that the site where the ACE-SPED is located is own by the University of Nigeria and does not need acquisition. In addition, the trees spotted within some portions of the land were non-economic trees. There were no settlements nor farms likely to be encroached upon during civil works. However, key activities in this phase will include vegetation clearing and mobilization of equipment and workers.

Heavy-duty vehicles, equipment, materials and workforce will be mobilized to the project site. Some of the construction equipment may include bulldozers, back hoes, chip spreaders, motor graders, roller machines, excavators, wheel loaders (pay loaders), pavement planers and compactors. Vegetation/site clearing will be limited to the area needed for the construction and will occur in phases to mitigate impacts of erosion, and land degradation.

Construction Phase
Key activities in this phase will include earthwork and grading, construction of buildings and drainage infrastructure as well as paving and landscaping. Construction works will essentially

entail civil engineering works requiring excavation, movement of earth materials, cement and concrete works, cuttings and fillings.

Operation and Maintenance Phase

Following construction, the buildings will become operational. The initial stage of the operation phase is the Defect Liability Period during which the buildings and ancillary infrastructures will be monitored and maintained by the contractors to ensure sustainability. There will be an annual repair, mechanized maintenance, and fumigation of facilities. Although computers and electronic devices will not be procured in large quantities, it is not inconceivable that e-waste will result from use of computers, CCTV and cameras which if not well disposed will cause environmental pollution and health hazards. Therefore, the impact of this and mitigation measures are addressed in this ESMP

2.2 Description of the Hostels Rehabilitation

The hostel rehabilitation project is for the two hostels out of four located in the Scholar village of UNN designated for the ACE-SPED Project. The hostels are dilapidated and are in need of renovation to make them much more befitting for students habitation. The figures below show the dilapidated structures in the hostel.

Fig. 3: One of the hostels mapped for rehabilitation

Fig.4: Another hostel mapped for rehabilitation

Activities for the Rehabilitation of Hostels

The rehabilitation works will include:

- scraping of washed off paints, screeding, scraping of dented floors; removal of roofing sheets, broken windows and doors,
- repairs of plumbing and electrical fittings.
- filling of cracked walls, fixing of ceiling, re-roofing, replacement of doors & windows, re-flooring, repainting; and
- upgrade of toilet facilities to modern toilets and provision of hygiene facilities.

CHAPTER 3: BASELINE DESCRIPTION OF PROJECT AREA

3.1 Introduction

The UNN ACE-SPED building and the site for Hostel Renovation are both located within the University of Nigeria, Nsukka and lies between longitudes 7°24'E and 7°26'E, and latitudes 6°51'N and 6°53'N. Nsukka is a town in Enugu state located in the South east geopolitical region of Nigeria. It is situated within Latitude 6.86°N and Longitude 7.39°E, 456 meters elevation above the sea level and has an area of 1810KM^2.

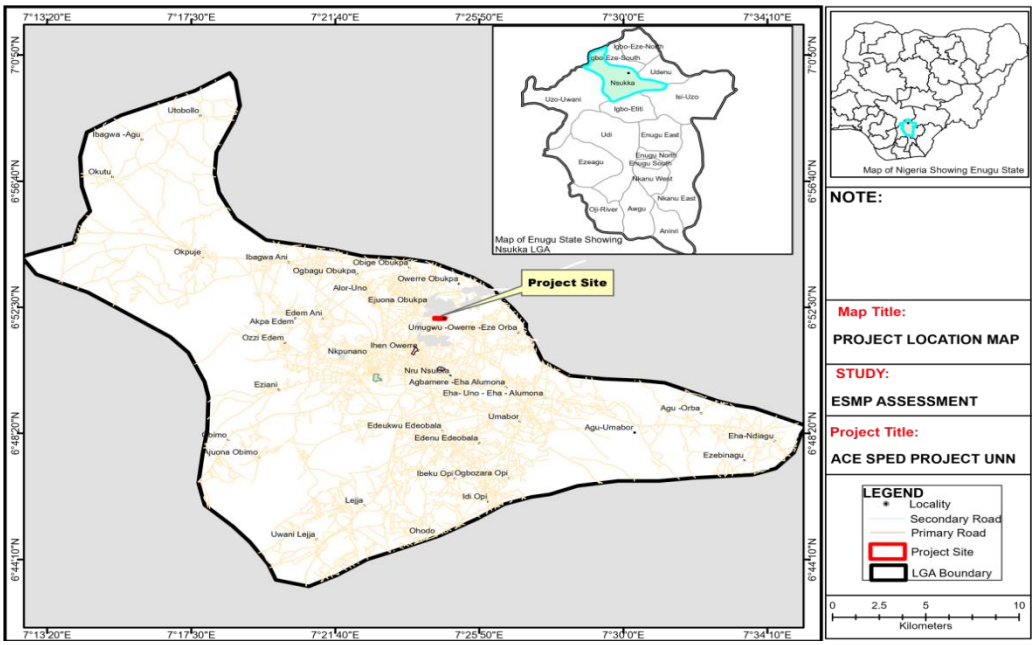


Figure 5: Map of Nsukka showing the project location

3.2 Description of Biophysical Environment of the Project Area

The environmental baseline situation in the project area was surveyed through desktop review, measurement of air quality, noise, soil and ground water as presented in this section.

3.2.1 Ambient Air quality and noise pollution measurement

The ambient air quality and noise baseline survey was measured at three different points in each of the two project sites using Aeroqual Series 300 hand held instrument for air pollutants and Extech Sound Level Meter 407732 for noise level measurement. Criteria pollutants measured are Suspended particulate matters (SPM), Carbon Monoxide (CO), Sulphur dioxide (SO_2), Hydrogen Sulphide (H_2S), Volatile Organic Compounds (VOC), Ammonia (CH4) and Nitrogen Dioxide (NO_2). Results of the ambient air and noise baselines are as follow:

Site: Scholars Village (Transit Camp)

Parameters	Location 1 (N6° 51' 54.2" E7° 24' 1.24")	Location 2 (N6° 51' 53.6" E7° 24' 1.0")	Location 3 (N6° 52' 12.8" E7° 24' 12.1")	WHO/FMEnv Limits
Noise (dB)	56.7	49.7	51.7	70-90dB
SO_2 (ppm)	0.00	0.01	0.00	2ppm
H_2S	0.00	0.00	0.00	20ppm
NO_2 (ppm)	0.091	0.09	0.093	0.8 ug/m3
CO (ppm)	0.00	0.00	0.00	10ppm
$PM_{2.5}$ (µg/m3)	43	35	32	250 ug/m3 daily
VOC (ppm)	0.09	0.06	0.05	0.75 ppm
CH_4	0.00	0.00	0.00	100ppm

Sources: UNN ACE-SPED ESMP Survey Team 2023; *Guidelines & Standards for Environmental Pollution Control in Nigeria (FEPA 1991). Air Quality Guideline (WHO, 2005)*

Site: ACE-SPED Project Site

Parameters	Location 1 (N6° 52' 7.02" E7° 24' 40.11")	Location 2 (N6° 52' 7.4" E7° 24' 40.85")	Location 3 (N6° 52' 6.4" E7° 24' 49.04")	WHO/FMEnv Limit
Noise (dB)	55.7	49.3	49.7	70-90dB
SO_2 (ppm)	0.00	0.00	0.00	2ppm
H_2S	0.00	0.00	0.00	20ppm
NO_2 (ppm)	0.088	0.1	0.09	0.8 µg/m3
CO (ppm)	0.00	0.00	0.00	10ppm
$PM_{2.5}$ (µg/m3)	29	28	27	250 µg/m3 daily
VOC (ppm)	0.04	0.05	0.05	0.75 ppm
CH_4	0.00	0.00	0.00	100ppm

Sources: UNN ACE-SPED ESMP Survey Team 2023; *Guidelines & Standards for Environmental Pollution Control in Nigeria (FEPA 1991). Air Quality Guideline (WHO, 2005)*

Discussion of Results of Air Quality and Noise Level Assessment

- Pollutants like Sulphur dioxide (SO_2), Hydrogen sulphide (H_2S), Carbon Monoxide (CO) and Ammonia (CH4) were not detected at the various sampling locations in the proposed ACE-SPED building. However, in the Scholar village (transit camp) being the site for hostel renovation the same results were applicable except for SO_2 which recorded 0.01ppm.

- PM recorded within the sampling locations ranged between 27– 29 ug/m3 at ACE-SPED site and between 32-43 ug/m3 at the Hostel renovation site. These values are within the 250 ug/m3 threshold of the WHO. VOC level recorded at ACE-SPED site ranged between 0.04 and 0.05ppm; while the results at the Scholar village hostel renovation site are between 0.05 and 0.09ppm. Also, these results were within the 0.75ppm threshold of the FMEnv and WHO. The results for Nitrogen dioxide in the ACE-SPED site ranged from 0.088 and 0.1ppm compared to the 0.09 and 0.093 recorded from the Scholar village hostel renovation area. The threshold for NO_2 according to WHO and FMEnv is 0.80ppm.

- The average day noise level was measured at three different points each for the ACE-SPED project site and the Scholar village hostel renovation sites. The lowest average result recorded

at the ACE-SPED site was 49.3 dB while the highest noise level recorded was 55.7dB. In the Scholar village Hostel renovation site the lowest noise level recorded was 49.7 and the highest noise level was 56.2dB. FMEnv permissible Noise Exposure Limits of 90 dB and the WHO (70 dB -A) permissible noise exposure limits.

The air quality assessment results indicate that all pollutants measured were either not detected or within acceptable FMEnv /WHO limits.

3.2.2 Baseline Ground Water Quality Conditions

On the premise that there is no surface water located within the project area of influence (up to 1.5km radius), the project relied on the sampled ground water from existing borehole (at latitude 6.864732 and longitude 7.418927) in the project area to ascertain the safety condition of drinking water in the area. Parameters measured and results are as follows:

Parameters	**GW**	**FMEnv Limit**
Physicochemical Properties		
Colour (TCU)	Colourless	15 (Colourless)
Temp ˚C	27	40
pH	7.0	7.5
Turbidity (NTU)	6.8	100
BOD (mg/L)	21.8	NA
TDS (mg/L)	86	1000
TSS (mg/L)	62.4	30
Salinity (mg/L)	28.75	NA
EC (µS/cm)	20.2	NA
Total Hardness (mg/L)	36.0	200
$NO_{3\ (mg/L)}$	3.8	10
$SO_{4\ (mg/L)}$	4.20	500
$PO_{4\ (mg/L)}$	0.30	5
Cl (mg/L)	18.6	0.4
DO (mg/L)	6.80	7.5
COD (mg/L)	20.4	80
BOD (mg/L)		
Heavy Metal Properties		
K (mg/L)	<0.01	NA
Ca (mg/L)	2.67	NA
Cu (mg/L)	0.00	0.1
Fe (mg/L)	0.85	1

Source: UNN ACE-SPED ESMP Survey Team, 2023

The physiochemical properties of the water were all within the permissible limit of the federal ministry of environment (FMEnv) except for Total suspended solid (TSS) and Chloride (Cl). Similarly, the results of the heavy metal properties shows that all the parameters measured were within the FMEnv allowable limits.

3.2.3 Baseline Soil Quality Conditions

Soil samples were collected at various points around the project area. Details of the collection points is presented below:

Sample Location	Sample Code	Coordinates	
		Latitude	Longitude
Topsoil	TS-01	6.868864	7.411369
Topsoil	TS-02	6.869128	7.413165
Topsoil	TS-03	6.868963	7.413578
Subsoil	SS-01	6.868864	7.411369
Subsoil	SS-02	6.869128	7.413165
Subsoil	SS-03	6.868963	7.413578

The Soil physicochemical properties measured were temperature, Ph, NO_3, SO_4, PO_4 and CL while Mn, Cu, Fe and Cd were measured for the heavy metals. Outcome of the survey is presented in the matrix below.

Parameters	Topsoil	Subsoil	FMEnv Limit
Physicochemical Properties			
Temp ˚C	28.00	27.00	40
pH (mol/L)	6.93	6.93	
$NO_{3\ (mg/L)}$	8.43	7.93	20
$SO_{4\ (mg/L)}$	4.93	4.40	500
$PO_{4\ (mg/L)}$	1.37	1.17	5
Cl (mg/L)	22.40	21.93	250
Heavy Metals			
Mn (mg/kg)	<0.09	<0.06	NA
Cu (mg/kg)	0.00	<0.00	1
Fe (mg/kg)	<82.77	73.24	1.5
Cd (mg/kg)	<0.07	0.04	0.8

Source: UNN ACE-SPED ESMP Survey Team, 2023

All the physicochemical properties were within the FMEnv allowable limits. Also, both top soil and sub-soil results show that the parameters were within the allowable threshold of the FMEnv except Fe which was above the limit of the FMEnv. The elevated result of the Fe (mg/kg) may be due to soil organic matter and pH condition precipitated by anthropogenic activities around the project site.

Outcome of other environmental parameters in the project area are summarized as follows:

Parameter	Description
Climate	Temperature typically varies from 610F to 97oF. Highest temperature occurs in February. Two climatic seasons occur in the project area: wet season (March to November) and dry season (December to February)
Rainfall	Rainy period of the year last for 9.4 months from March to November with average rainfall of between 2000 to 2500mm
Vegetation and soil	The study area lies within the tropical rainforest/Guinea Savannah belt (Iloeje, 1978). The soil is classified as the rain forest and lateritic soils, and are highly leached by heavy rain fall
Topography	Nsukka has maximum elevation of 580m, average elevation of 319m and minimum elevation of 65m

3.3 Description of Socio-economic baseline Conditions of the Project Area

The social baseline of the project area is summarized in the Table below.

Variables	Description
Religion	Predominantly Christian with slam, and other religion.
Population	The university community has an estimated population of over 20000 residents (but residents of Eni-Njoku street (staff quarter), location of the project are less than 40 people). The students for ACE-SPED is estimated at 2000 persons with a gender ratio of 0.48: 0.52 for male and females respectively
Ethnicity and Language	University of Nigeria community is relatively heterogeneous in terms of ethnicity and language but more of Igbo. The ethnic groups include Igbo, Yoruba, Hausa, Fulani, Tiv, Igala, and other groups
Migrant Status	Migrants from other ethnic groups settle in the community and are engaged in teaching, administrative, and learning activities.
Educational Institutions	They include: University of Nigeria, Nsukka, University primary and secondary school.
Source of drinking water	Piped water within the University community as well as Sachet water from University Lion Water
Housing	Houses are built with cement blocks and corrugated iron roofing sheet.
Access Road to and within Community	The roads both leading to the community and within the community are tarred.
Access to Electricity	The area is connected to the national grid but some of the residents support their power supply with solar energy and generators for power
Access to telecommunication	The area is connected to network service providers such as MTN, GLO, and Airtel
Access to Market	There is a nearby market that is within 1km or 10 minutes driving distance from the University campus-Ogige market. There are other commodity markets that are within 2km to 5km from the University community such as the Orba market and Obollo Afor Market.
Means of Transportation	The common forms of transportation within and out of the community are commercial taxi and private vehicles
Channel of Information	Information within the University is disseminated through the school hierarchical structure (VC, head of faculty, head of department, head of students affairs and head of students union). Information within adjacent community to the University is made available through youth leader, town crier and village messenger

CHAPTER 4: ASSESSMENT OF POTENTIAL ADVERSE ENVIRONMENTAL AND SOCIAL IMPACTS

4.1 Introduction

Impacts of the ACE-SPED project includes positive and adverse impacts. Positive impacts were assessed through consultation with potential beneficiaries and through review of ACE-SPED project documents. Similarly, adverse impacts of the ACE-SPED project identified in the previous chapter were assessed and quantified using the Rank Assessment Matrix with result as presented in the matrix below.

Aspect	Potential Impacts	Probability	Consequence	Risk Ranking	Control
Air quality	Deterioration of local air quality through dust generated from construction and renovation works	likely	moderate	Medium	See ESMP Table section
Noise and vibration	Temporary elevated noise level during construction, installation and renovation activities	possible	moderate	Medium	
Soil quality	Contamination of soil and plant nutrients due to leakage and spills from oils and lubricants used by the heavy equipment and trucks during construction stage	possible	Moderate	Medium	
Waste management and pollution	Potential contamination of land and ground water due to poor handling of wastes	Possible	Significant	Medium	
Occupational health and safety	Injuries and deterioration of health due to non-adherence to industry OHS guidelines	Likely	Moderate	Medium	
Insecurity	Security concerns as a result of influx of workers and vendors	Possible	Moderate	Low	
GBV/SEA	Incidences of GBV may arise from the interaction of men folks with female students and staff	Possible	Significant	Medium	
Grievances	There may be grievances between the client and contractors and between workers and employers, etc as a result of the project	possible	moderate	Low	

4.2 Positive Impacts of the Project

The project is expected to enhance impactful educational, research, development and training activities in Electric power systems development; Renewable energy, waste-to-energy and energy conservation; Energy resources assessment and forecasting; Sustainable energy materials and Energy Policy, Regulation and Management.

It will contribute to the overall socioeconomic well-being of the benefiting host community through the provision of employment opportunities. During construction, some locals are expected to be engaged directly especially as semi-skilled (e.g. masons, carpenters, electrical technicians, welders, etc.) and unskilled labour (e.g. for site clearance, loading and offloading of materials, security services etc.). There will also be indirect opportunities for local contractors, businesses and food sellers amongst others. This will be a significant impact since the rate of unemployment is generally high in Nigeria and in adjoining project communities.

4.3 Adverse Environmental Impacts

Inevitably, the project will have some minor adverse impacts on the biophysical and social environment. The minor environmental and social impacts are expected to be typically more pronounced during the construction phase of the project. These impacts will be largely localized and short term, and can be avoided or managed through the application of appropriate mitigation measures, sound design, good construction practices, effective maintenance and adequate supervision and enforcement during project implementation. The description of the potential environmental and social impacts that may arise from activities to be undertaken during the preconstruction, construction and operation phases of the proposed project are discussed in the following section.

Deterioration of Local Air Quality

The project will inevitably lead to an increase in the release of gaseous pollutants and dust to the atmosphere within the vicinity of the project corridor. The major sources of air pollution during the project will include construction activities and movement of vehicles. Activities likely to pose adverse impact on air quality will include site clearing, earthworks, movement of vehicles on unpaved surfaces and transportation of road building materials.

Noise and Vibration

Noise is usually associated with construction especially building development projects. During pre-construction and construction, major noise sources will be from the operation of heavy-duty vehicles and motorized equipment and machineries such as concrete mixers etc. These sources of construction noise could pose significant risks to construction workers and nuisance to community members.

Soil Erosion

Ground disturbing activities during preconstruction and construction phases will inevitably expose top soil to erosion. Activities such as site clearing with removal of vegetative cover, compaction, excavation, side tipping of spoil materials could be responsible for this.

Soil Contamination

Accidental leakages or spills of hazardous materials such as fuels, oils and lubricants used in operating and maintaining vehicles and equipment as well as from their improper handling and storage may cause direct contamination to soil especially during construction and operation and maintenance phases of the project. Indiscriminate disposal of hazardous wastes during preconstruction and construction phases may also lead to significant contamination of soils.

Waste generation Impact

Waste streams from the project will come from cleared vegetal materials during pre-construction phase, debris and spoils from rehabilitation and construction of the lecture theatre, conference rooms and workshops, as well as e-waste from the use of computers, and CCTV camera.

4.4 Adverse Social Impacts

Impacts of Land Acquisition

Land acquisition is not envisaged in this project because the land for which the project will be sited is own by the University of Nigeria, Nsukka. Also, there are no economic crops and private assets to be affected, thus there will be no involuntary resettlement.

Threat to University community safety and security

Influx of migrant workers and business opportunists who arrive at the project site to work or do business during construction may escalate crime rate and insecurity including kidnapping and armed robbery.

Risk of Sexually Transmitted Diseases

There is the potential for increased Sexually Transmitted Diseases (STDs), particularly during pre-construction and construction phases of the project, due to increased concentration of people such as construction workers, migrant workers and camp followers with diverse cultural and social backgrounds. Inevitable interactions among these migrant population and students may cause increased prevalence of STDs especially HIV/AIDS.

Impact of vector diseases

Reducing the impacts of vector borne diseases shall be the responsibility of the contractor and the supervising consultant will ensure adherence to this requirement. This should be accomplished through implementation of diverse interventions aimed at eliminating the factors that lead to vector disease including: i) prevention of larval and adult propagation of vectors through sanitary improvements and elimination of breeding habitat close to human settlements and, ii) eliminate any unusable impounding of water.

Security Challenges

Security challenges may arise in the project area due to the storage of equipment and materials on site. Storage of these materials may attract thieves and hoodlums to the area with increased risks of threats to lives and properties. To prevent security challenges, trained security personnel shall be deployed to secure the site and main entrance gate scaling up proactive profiling of movement of persons and vehicles during construction phase.

Impact on Occupational Health and Safety

Occupational health and safety issues will likely be of concern particularly during pre-construction and construction phases of the proposed project. Activities such as mobilization of equipment and materials, site clearing and demolition, working on heights, etc may increase construction workers exposure to occupational health and safety hazards. Consequently, there may be potential risks to workers' health and safety with increased predisposition to accidents and injuries.

Labour Influx and Gender Based Violence

This sub-section discuses some key potential social impacts which requires great deal of attention and therefore, must be prioritized and mitigated during project implementation. This does not undermine the need to mitigate other social impacts identified above. These key social areas are labour influx, gender-based violence and HIV/AIDS as discussed below:

- Labor Influx

During project implementation, the arrival of construction workers and business vendors may result in some labor influx-related risks such as workers' sexual relations with female students which may result to pregnancies, the spread of HIV/AIDS and sexual harassment. To ensure adequate protection of students and UNN community against the vices highlighted above, the

project management (PIU) should ensure the implementation of the following additional mitigation measures:

(a) establishing and enforcing a mandatory Code of Conduct for the contractor, managers and workers, and an Action Plan for implementation;

(b) taking countermeasures - indicated in the ESMP - to reduce the impact of the labor influx on public services

- **Gender Based Violence**

GBV risks in the project areas might include Intimate Partner Violence (IPV), public harassment including verbal insults, physical abuse, and rape. Development and implementation of specific GBV risk prevention and mitigation strategies, tailored to local contexts, will be critical. Therefore, project implementation unit will include in the bidding documents ('pre-qualification' and 'employers' requirements') key principles and specific requirements to address GBV so as to reduce and mitigate the risks of GBV especially during project implementation. Such measures will include:

- GBV/SEA assessment of project;
- ✓ Mandatory contractors' code of conduct on sexual harassment;
- ✓ Monthly site visit by the safeguard unit/GBV officer to monitor GBV/SEA during construction/implementation phase;
- ✓ Workers' sensitization on GBV/SEA;
- ✓ Provision of referral units for survivors of GBV/SEA;
- ✓ Provisions in contracts for dedicated payments to contractors for GBV/SEA prevention activities against evidence of completion;

The following actions are recommended for immediate implementation by ACE-SPED implementation unit:

- ✓ Strengthen/train the safeguards unit to handle issues on GBV/SEA and to manage social risks associated with GBV/SEA in the project.
- ✓ Address risks of GBV/SEA by providing guidance, awareness, and dissemination of relevant GBV/SEA materials to the people living in the project area including students
- ✓ Develop a clear project specific internal "Reporting and Response Protocol" to guide relevant stakeholders in case of GBV/SEA incidents,
- ✓ Identify development/stakeholders and cultivate pragmatic partnership on GBV/SEA prevention measures and referral services,
- ✓ Develop Codes of Conduct for civil works contractors with prohibitions against GBV/SEA,
- ✓ Strengthen consultations and operationalizing GBV/SEA specific GRM,
- ✓ Provide financial support for implementation of the GBV/SEA actions described herein, including training and awareness building for various stakeholders,
- ✓ Provide toe free lines to key community members and maintenance groups for reporting project related social complain

CHAPTER 5: ENVIRONMENTAL AND SOCIAL MANAGEMENT PLAN (ESMP)

5.1 Introduction

This section is the core of this ESMP. It specifies all identified impacts at the different stages of the sub-projects, proffers specific mitigation measures, responsibility for mitigation and cost of mitigation. The section also unbundles monitoring plans to ensure effective project implementation. The monitoring plan describes parameters to be monitored, method of measurement, performance indicators, monitoring frequency and cost of monitoring. All these are embedded in the ESMP table prepared for the ACE-SPED construction and the hostel renovation sub-projects separately for use of the project team and contractors.

5.2 Environmental and Social Impacts, Mitigation and Monitoring Programme for the ACE-SPED Building

Below (Table 5.1) are the highlights of the positive and negative impacts of the project which are to be enhanced/mitigated through the environmental and social management plan.

Table 5.1: Potential Positive and Negative Impacts of the Project

Positive Impacts	
- The project will enhance impactful educational, research, development and training activities in Electric power systems development; Renewable energy, waste-to-energy and energy conservation; Energy resources assessment and forecasting; Sustainable energy materials and Energy Policy, Regulation and Management. - It will contribute to the overall socioeconomic well-being of the benefiting host community through the provision of employment opportunities. - During construction, some locals will be engaged as semi-skilled workers (e.g. masons, carpenters, electrical technicians, welders, etc.) - There will also be indirect opportunities for local contractors, businesses and food sellers amongst others. - Improved research staff capacity to engage in quality and novel research	
Adverse Impacts	
Environmental Impacts	**Social Impacts**
- Increase in fugitive dust emission during pre-construction and construction phases of the rehabilitation. - Waste generation including vegetation waste, construction wastes, municipal wastes and e-wastes - Possibility of seepage of fuel from machineries and effluent discharge into	- Noise and vibration from heavy-duty project vehicles and equipment resulting in nuisance to residents and students. - Loss of employment for labourers after the completion of the construction and rehabilitation works. - Disruption of traffic movement. - Occupational Health and Safety risks.

the environment, thus impacting on the environmental quality. - Loss of species diversity and abundance, including soil organisms, fungi, invertebrates, and bacteria due to site clearing - Air quality deterioration	- Spread of sexually transmitted diseases (HIV, HB, Gonorrhea, etc.) - Increase in insecurity as a result of influx of migrant workers - Gender-based violence and sexual molestation may result from interaction of construction staff, migrant workers and female students

The potential E&S impacts likely to result from the interaction between the project activities in each phase and the environment as well as comprehensive mitigation measures are presented in the ESMP table.

5.2.1 Environmental and Social Management Table for the ACE-SPED Building

The ESMP Table highlights activities, identified adverse impacts, mitigation Measures and corresponding Indicator(s), Mode of Measurement, corresponding Cost of Mitigation, monitoring indicators, Frequency, Monitoring Cost, as well as responsibilities for implementing these measures.

Table 5.2: ESMP Table of Impact, Mitigation Measures and Implementation and Monitoring for the ACE-SPED Building

	Potential Impact	Mitigation Measures	Responsibility for Mitigation	Mitigation Cost (Naira)	Parameters to be measured	Method of Measurement	Performance Indicator	Sampling Location	Monitoring Frequency	Responsibility for Monitoring	Monitoring Cost (Naira)
rA	**Pre-construction Phase**										
Activity: Site Clearing & Mobilization											
Environmental Impacts											
1	Loss of vegetation from site clearing	Clear only areas that are necessary Revegetate unused cleared areas	Contractor	220,000	Loss of vegetation	Visual observation	Contractors compliance	Within the school	One off	ESO, PIU	Part of routine monitoring
2	Soil erosion and soil compaction due to movement stacking of heavy-duty equipment	Limit zone of vehicle and equipment weight impacts (designate an area for parking and stacking equipment)	Contractor	40,000	Visible demarcation of vehicles and equipment limit zone	vehicles and equipment limit zone Visual observation Soil Compaction test	Contractors compliance	Project camp sites and equipment packing zones	Monthly	ESO, PIU	80,000
3	Soil pollution from leakages of stacked equipment and chemical substances into soil.	Fasten loose parts (bolts, nuts); Install impermeable surface at the limit zone to contain	Contractor	80,000	Installation of impermeable platform at limit zone.	Soil quality test Site inspection	FMEnv soil pollutants permissible limit	Project camp sites and equipment packing zones	Monthly	ESO, PIU ESEPA	

		potential leakages									
4	Air quality deterioration: fugitive dust, exhaust fumes, GHGs	Sprinkle water via spraying devices to limit dusts.	Contractor	120,000	Gaseous pollutants: SO2, NO2, CO2, CO, VOCs, H2S, TSP	In-situ Air Quality Measurement Visual Observation	FMEnv air pollutants permissible limit	1 km radius of project area	Weekly	ENMOE, ESEPA, Environment Safeguard Officer (ESO) PIU	100,000
		Service vehicles; undergo vehicle emission testing (VET) and vehicle exhaust screening	Contractor	120,000 (@ N40,000/service x 3)	VET/VES Report		FMEnv air pollutants permissible limit	On-site and nearby community	Every two months		

Activities	Potential Impact	Mitigation Measures	Responsibility for Mitigation	Mitigation Cost	Parameters to be Measured	Method of Measurement	Performance Indicator	Sampling Location	Monitoring frequency	Responsibility Monitoring	Cost (Naira)
A	PRE-CONSTRUCTION PHASE										
Site clearing and Mobilization											
Environmental Impact											
5	Noise	Avoid work at night Retrofit equipment with sound proof	Contractor	- 120,000	Number and frequency of complaints in project area	In-situ measurement of noise level	Noise level not to exceed 90dB(A) for 8 hours working period	1 Km radius of project site	Weekly	ESO, PIU ENEPA	100,000
Social Impacts											
6	Occupational health risks	Provide PPE (Hardhat, hand gloves, safety boots, .) to workers; Educate site workers; Provide first aid onsite	Contractor PIU	300,000 (for PPEs), 200,000 (for first aid kit)	Number of incidents / accidents	Incident/ accident report Routine inspection	Use of PPEs Contractors compliance	ACE-SPED site premises	Weekly	PIU	100,000
7	Noise	Retrofit equipment with sound proof	Contractor	160,000	No. and frequency of complaints in project area	In-situ measurement of noise level	Noise level below 90dB(A)	1 Km radius of project site	Weekly	SSO, PIU ENEPA	See A3

8	Public Safety	Provide caution/warning signs at sensitive locations Use trained drivers	Contractor	See A6 above	Number of incidents/ accidents	Incident/ accident report Routine inspection	Contractors compliance Caution signs	Routes through community to UNN	Every 2 weeks	UNN Management SSO, PIU	See A5 above
Sub-Total				**1,360,000**							**380,000**
Activities	Potential Impact	Mititgation Measures	Responsibility for Mitigation	Miigation Cost	Parameters to be Measured	Method of Measurement	Performance Indicator	Sampling Location	Monitoring Frequency	Responsibility(Monitoring)	Cost (USD
B	**CONSTRUCTION PHASE**										
Operation and movement of equipment											
Environmental Impact											
1	Open defecation by workers	Provide stationery toilets for workers to be handed over to the School afterwards/ Sensitize workers against open defecation	Contractor	800,000	Citing of faecal waste within the site	Visual observation	Absence of faecal waste onsite	Site Area	Weekly	PIU	Part of routine monitoring
2	Air quality deterioration from exhaust fumes, hazardous gases (NOx, CO,	Service equipment and machinery regularly	Contractor	120,000 (@ N40,000/service x 3)	Gaseous pollutants: SO2,	In-situ Air Quality Measurement	FMEnv air pollutants permissible limit	Site Area	Every two months	ENMOE ENEPA	120,000

	SOx, SPM,), Oxides from machinery operations Climate Change: Green House Gas Emissions	Turn off machines when not in use			NO2, CO2, CO, VOCs, H2S, TSP	Visual Observation VET/VES Report					
3	Soil and groundwater contamination by oil spills, lubricants and other chemicals	Site oil and lubricants on an impervious base and should have drip pans Locate storage area far from boreholes, all containers should be clearly labeled	Contractor	150,000	Soil quality parameters Compliance with fuel storage procedures	In situ/ laboratory analysis Visual/ Observation	FMEnv soil pollutants permissible limit	Site Area	Twice during construction	ESO, PIU ENEPA	160,000
4	Pollution from presence of construction waste on-site	Implement site-specific waste management plan Liaise with ENEPA for effective waste management and safe handling/disposal of waste	Contractor	500,000	Presence of construction waste on-site	Site Inspection	Compliance with waste management plan Good house keeping	Site Area	Weekly	PIU, ESO, ENEPA	200,000

Activity	Potential Impact	Mitigation Measures	Responsibility for Mitigation	Mitigation Cost	Parameters to be Measured	Method of Measurement	Performance Indicator	Sampling Location	Monitoring Frequency	Responsibility (Monitoring)	Cost (USD)
B	CONSTRUCTION PHASE										
Operation and movement of equipment/ Demolition/ Rehabilitation works											
Social Impact											
6	Respiratory infections from air contamination by exhaust fumes and dust	Regular sprinkling of water during construction works. Distribute face masks to PAPs	Contractor	200,000	Prevalence of respiratory infections reported in the community around the site Area	Rapid health survey	Availability of face masks on site	Site Area	Weekly	School Mgt. PIU	100,000
7	Security vulnerability as a result of influx of migrant workers during construction work	Work with the UNN security unit to ensure profiling of all movement into the site/school	Contractor	200,000	No of complaints received within the construction period	Consultation	Minimal number of reported cases	Routes through community to the site	Weekly	School Management Security Unit, , PIU	Part of routine monitoring

8	Labor Influx: Potential risk of Sexual Exploitation and Abuse (SEA)/ Gender Based Violence (GBV)	Contractor workers will sign Code of Conduct; Sensitize workers on zero tolerance for sexual assault	Contractor	part of OHS training by contractor	Stakeholders concerns on risk of GBV	Consultations GBV Incident Report	Signed CoCs with the PIU Conduct of sensitization campaigns	University community	Twice during Construction	SSO, PIU	100,000
9	Labor influx3 : Increase in spread of STDs/STIs in the School	Sensitize workers on sexual diseases, and distribute condoms to contractor staff	Contractor	100,000	Level of Awareness and Education No of new STI cases	Rapid health survey	Level of awareness on preventive measures. % of reported STI/ STD cases	Project area of influence Health care	Twice during Construction	SSO, PIU Primary Health Care Workers	50,000
10	Conflicts between contractors, Residents and school,	Good work enforcement program Grievance Redress Mechanism Regular consultations	Contractor GRC	500,000	No of complaints received	Consultation Review Grievance redress Log	No of cases Resolved by the GRC	School premises Project area of influence	Continuous	GRC	100,000
11	Vehicular or on-site accident	implement the prepared traffic management plan Ensure the installation of speed limit signage and speed	Contractor	200,000	No of reported accidents	Reporting and feedback mechanism Accident Report	Zero/ minimal accident reports	School premises Project area of influence	Continuous	ESO, PIU Security Unit of	100,000

		breakers									
12	Risk of occupational accidents, injuries and diseases	Implement project specific HSE Plan Provide and enforce usage of appropriate PPE.	Contractor	200,000	HSE Plan No of reported cases	Accident Reports	Compliance with HSE Plan	School Premises Project area of influence	Monthly	ESO, PIU SSO,	100,000
13	**Employment:** Temporal jobs created and skill gained may be lost at project closure	Provide Testimonial or Certificate to local workers and all that were trained by the Project to help them secure employment at the end of ACE-SPED	PIU	-	Certificates or Testimonials issued out	Interview of the local employees and trainees	No of persons that received Testimonials or Certificates	Project area of influence	Annual	PIU	NA
Sub-Total			**2,970,000**								**1,030,000**

Activity	Potential Impact	Mitigation Measures	Responsibility for Mitigation	Mitigation Cost	Parameters to be Measured	Method of Measurement	Performance Indicator	Sampling Location	Monitoring Frequency	Responsibility (Monitoring)	Cost (USD)
C	OPERATION PHASE										
Demobilization of equipment and construction materials from the school											
1	Soil and water contamination from oil/ lubricants	Cart away all spoils through the relevant authorities Clean out impact areas	Contractor	300,000	Oil Spillages, construction waste and spoilt equipment/parts	Site inspection	Good house keeping	Hostel, workshop and laboratory premises and Campsite	Quarterly	ESO, PIU	100,000
2	Waste generation	Implement on-site waste management plan	UNN Mgt	Part of yearly budget	Air, water and soil pollutants	pollutants In-situ air, water and soil sampling	Good waste management practice	Site Areas	As required	ENEPA	Part of annual budget
Demobilization of equipment and construction materials from the site											
Social Impacts											
1	Disruption in learning activities.	Limit demobilization activities to non-school periods	Contractor	-	No of complaints from beneficiary students	Interview	Stakeholders satisfaction with activity	Site Areas	As required during	UNN mgt	
Sub-Total				**300,000**							**100,000**
Total				**N4,510,000**						**N1,590,000**	

5.3 Environmental and Social Impacts, Mitigation and Monitoring Programme for the Hostel Renovation

Major environmental and social concerns for the hostel renovation includes accommodation for students, who were staying in the hostels, wastes generation and disposal, air and noise pollution, health and safety of workers and public as well as STDs and GBV/SEA risks.

Table 5.3: ESMP Table of Impact, Mitigation Measures and Implementation and Monitoring for the Hostel Renovation

	Potential Impact	Mitigation Measures	Responsibility for Mitigation	Mitigation Cost (Naira)	Parameters to be measured	Method of Measurement	Performance Indicator	Sampling Location	Monitoring Frequency	Responsibility for Monitoring	Monitoring Cost (Naira)
A	RENOVATION PHASE										
Activity: Mobilization of materials and renovation works											
Environmental Impacts											
1	Pollution from Waste materials	Implement site-specific waste management plan (annex 7) Liaise with OGEPA for effective waste management and safe handling/disposal of waste	contractor	500,000	Presence of waste materials	Site inspection	Compliance with waste management plan	Project site	weekly	ESO, ENSWAMA	300,000
2	Air quality deterioration due exhaust fumes from vehicles transporting materials to site	Service equipment and machinery regularly Turn off machines when not in use	Contractor	120,000 (@ N40,000/service x 3)	Gaseous pollutants: SO2, NO2, CO2, CO, VOCs, H2S, TSP	In-situ Air Quality Measurement Visual Observation	FMEnv air pollutants permissible limit	Site Area	Every two months	ENMOE ENEPA	120,000

						VET/VES Report					
A											
Mobilization of Materials and Renovation Work											
Social Impacts											
1	Occupational health risks such as fall from height, injuries from use of tools and equipment	Provide PPE (Hardhat, hand gloves, safety boots, nose masks etc.) to workers; Educate site workers; Provide first aid onsite	Contractor PIU	300,000 (for PPEs), 200,000 (for first aid kit)	Number of incidents / accidents	Incident/ accident report Routine inspection	Use of PPEs Contractors compliance	ACE-SPED site premises	Weekly	PIU	100,000
2	Respiratory infections from air contamination and dust	Workers at the site should use face masks	Contractor	200,000	Prevalence of respiratory infections reported in the community around the site Area	Rapid health survey	Availability of face masks on site	Site Area	Weekly	School Mgt. PIU	100,000
3	Public Safety risks due to flying hazardous materials from the site	Cordon off the area during renovation	Contractor	300,000	Number of incidents/ accidents	Incident/ accident report Routine inspection	Contractors compliance Caution signs	Routes through community to UNN	Every 2 weeks	UNN Management SSO, PIU	100,000

4	Concern about students accommodation during the hostel renovation	During the renovation work, the students will be moved to other hostels that are not part of the renovation	Contractor	-	No of complaints by students	Consultations Review grievance log	Minimal number of reported cases	School Premises	Every 2 weeks	School Management ESO, PIU	Part of routine monitoring
5	Security escalation as a result of influx of migrant workers during construction work	Work with the UNN security network to ensure necessary profiling of all movement into the site/school	Contractor	200,000	No of complaints received within the construction period	Consultation	Minimal number of reported cases	Routes through community to the site	Weekly	School Management Security Unit, , PIU	Part of routine monitoring
6	Labor Influx: Potential risk of Sexual Exploitation and Abuse / Gender Based Violence (GBV/SEA)	All contractors workers to sign Code of Conduct; Sensitize workers on zero tolerance for GBV/SEA	Contractor	part of OHS training by contractor	Stakeholders concerns on risk of GBV	Consultations GBV Incident Report	Signed CoCs with the PIU Conduct of sensitization campaigns	University community	Twice during Construction	SSO, PIU	100,000
7	Increase in spread of STDs/STIs in the School	Sensitize workers on sexual diseases, and Distribute condoms to contractor staff	Contractor	100,000	Level of Awareness and Education No of new STI cases	Rapid health survey	Level of awareness on preventive measures. % of reported STI/ STD cases	Project area of influence Health care	Twice during Construction	SSO, PIU Primary Health Care Workers	50,000

8	Conflicts between contractors, Residents and school,	Good work enforcement program Grievance Redress Mechanism Regular consultations	Contractor GRC	500,000	No of complaints received	Consultation Review Grievance redress Log	No of cases Resolved by the GRC	School premises Project area of influence	Continous	GRC	100,000
9	**Employment:** Temporal jobs created and skill gained may be lost at project closure	Provide Testimonial or Certificate to local workers and all that were trained by the Project to help them secure employment at the end of ACE-SPED	PIU	-	Certificates or Testimonials issued out	Interview of the local employees and trainees	No of persons that received Testimonials or Certificates	Project area of influence	Annual	PIU	NA
Sub-Total				**2,420,000**							**970,000**

Activity	Potential Impact	Mitigation Measures	Responsibility for Mitigation	Mitigation Cost	Parameters to be Measured	Method of Measurement	Performance Indicator	Sampling Location	Monitoring Frequency	Responsibility (Monitoring)	Cost (USD)
C	OPERATION PHASE/DEMOBILIZATION OF EQUIPMENT										
1	Waste generation	Implement on-site waste management plan on a continuous basis	UNN Mgt	300,000	Air, water and soil pollutants	pollutants In-situ air, water and soil sampling	Good waste management practice	Site Areas	As required	ENEPA	100,000
2	Occupational health risks	Use PPes	Contractor PIU	As provided in A1	Number of incidents/ accidents	Incident/ accident report Routine inspection	Use of PPEs Contractors compliance	ACE-SPED site premises	Weekly	PIU	100,000
Sub-Total				**300,000**							**200,000**
Total				**N2,720,000**						**N1,170,000**	

5.4 ESMP Capacity Building Plan and Cost

It is critical that the project officers undergo training on ESMP implementation. This is imperative to build capacity and to be acquainted with the World Bank safeguard policies including GBV/SEA, grievance redress mechanism and HSE/OHS training. Table 5.4 presents the capacity building plans and estimated budget.

Table 5.4: Capacity Enhancement Plan and Cost

Item	Responsibility	Participants	Cost (Naira)
Training on ESMP Implementation: impacts, mitigation, resource efficiency and sustainable procurement roles & responsibilities	PIU/consultant	ESO, Project engineer and M&E Officers	500,000
Training of Program Action Plan (PAP) implementation	PIU/consultant	ESO, Project engineer and M&E Officers	500,000
Training on: ✓ Operationalizing GRM ✓ OHS ✓ Code of Conduct against GBV/SEA/SH	FPIU/consultant	ESO, Project engineer and M&E Officers	500,000
		Total	**N1,500,000**

5.5 ESMP Implementation Budget

The total estimated cost for the ESMP implementation and monitoring is **N13,936,000,** out of which N7,230,000 will be embedded in the contractors contracts, while the remaining **N6,706,000.00 (US$ 14,902)** is the component to be set aside for PIU ESMP implementation and monitoring**.** The breakdown of the estimated budget is presented in Table 5.5

Table 5.5: Breakdown of Budget for ESMP Implementation

Item	Responsibility	Cost Estimate in Nigerian Naira (N)	Cost Estimate in US Dollars (US$)*
Mitigation: ACE SPED Building construction	Contractor	Part of construction contract cost (N4,510,000)	
Mitigation: Hostels Renovation	Contractor	Part of renovation contract cost (N2,720,000)	
Monitoring (ACE-SPED Building and Hostel Renovation)	PIU	2,760,000	6,133
Capacity Building	SPIU, ENSMEnv and other relevant MDAs	1,500,000	3,333
GRM (10% of cost of total mitigation	PIU	**723,000**	**1,606**
GBV/SEA (10% of cost of total mitigation	PIU	**723,000**	**1,606**
Disclosure	PIU/FMEnv	1,000,000	2,222
Total		**6,706,000**	**14,902**

*1 US$ =N450

5.6 Institutional Arrangement for ESMP and Monitoring Plan

The roles and responsibilities of the different stakeholders are as contained in table 5.6 below:

Table 5.6: The Roles and Responsibilities of the Stakeholders in the Implementation, Monitoring and Review of the ESMP

S/N	Stakeholder	Roles and Responsibilities
1	ACE-SPED/PIU	● Liaise closely with Enugu State Ministry of Environment in preparing a coordinated response on the environmental and social aspects of project development respectively as required; ● Safeguards due diligence ● In case of any violations on arising works, PIU will request contractors to amend and correct the violation. ● Receive and supervise the environmental report from the Independent Environmental Consultant (IEC), ● Project implementation unit Safeguard Specialist will be in charge of review of environmental report and recommend further actions. ● Cooperate with WB to periodically supervise contractors' activities. ● Weekly meetings will be held between Contractors, PIU, and consultants. In addition, PIU is to have assigned personnel for regular inspection.
2	Enugu State Ministry of Environment	● Environmental compliance Coordinator at the State level ● Lead role - provision of advice on screening, scoping, review of draft ESMP report (in liaison with Federal Ministry of Environment) ● Site assessment and monitoring of ESMP implementation
3	Federal Ministry of Environment	● Lead role - provision of advice on screening, scoping, review of draft ESMP report (in liaison with State Ministry of Environment), ● Receiving comments from stakeholders, public hearing/ review of the project proposals, and convening a technical decision-making panel arising from the public disclosures, ● Project categorization for EA, Applicable standards, Environmental and social liability investigations, Monitoring and evaluation process and criteria
4	Other MDAs	● Other MDAs come in as at when relevant areas or resources under their jurisdiction or management are likely to be affected by or implicated by projects. ● They participate in the EA processes and in project decision-making that helps prevent or minimize environmental and social impacts and to mitigate them. These institutions may also be required, to issue a consent or approval for an aspect of a project; allow an area to be included in a project; or allow impact to a certain extent or impose restrictions or conditions, monitoring responsibility or supervisory oversight
5	AAU	● Association of African Universities will be responsible for implementation support of components 1 &2 and for the overall regional facilitation of ACE impacts
6	World Bank (WB)	● Overall supervision and provision of technical support and guidance. ● Recommend additional measures for strengthening the management framework and implementation performance; ● Supervising the application and recommendations of sub- project ESMPs.

7	Safeguards Unit	**Environmental Aspects** ● Collate environmental baseline data on relevant environmental characteristics of the selected project sites; ● Ensure that project activities are implemented in accordance to best practices and guidelines set out in the site specifics; ● Identify and liaise with all stakeholders involved in environment related issues in the project; and ● Be responsible for the overall monitoring of mitigation measures and the impacts of the project during implementation. ● Ensure the operationalization of Grievance Redress Mechanism (GRM) ● Ensure project beneficiaries and host communities are sensitized about the available reporting channels and how to access them ● Periodically monitor the GRM to ensure it is effective and fit for purpose ● Lead the process of disclosure **Social Aspects** ● Develop, coordinate and ensures the implementation of the social aspects of the ESMP; ● Implement the grievance mechanism on the project ● Identify and liaise with all stakeholders involved in social related issues in the project; ● Conduct impact evaluation and beneficiaries' assessment
8	Works Contractor	● Contractors must fully carry out the measures of the environmental protection, health and safety procedure as indicated; ● Any changes related to Environment, health and safety procedure must be informed to PIU for approval ● In case of any violations on arising works either detected by Environmental consultant, Safeguard Specialists, or new arising works proposed by contractors, they must be reported to PIU for further actions. ● If contractors decided to not follow instruction from PIU, construction activities will be halted until necessary actions are taken. ● Contractors must assign at site, personnel in charge of traffic management, Environment, health and safety procedure. ● Compliance to BOQ specification in procurement of material and construction
9	Site Engineer	● Provide oversight function during construction and decommissioning

5.7 Implementation Schedule

The activities related to environmental management and monitoring have to be integrated in the overall construction and rehabilitation schedule. The project implementation phase is estimated to be completed in 15 months. The implementation schedule is presented in Table 5.7

Table 5.7: ESMP Implementation Schedul**e**

S/N	Activity Description	Responsible	Preconstrution (Month)					Constructiuon (Month)									Operation		
			1	2	3	4	5	1	2	3	4	5	6	7	8	9			
15.	Clearance and Formal Disclosure of ESMP	PIU																	
16.	Inclusion of Environmental & Social Requirements in Bid Docs	PIU																	
17.	Allocating Budget for ESMP	PIU																	
18.	Appointing Support Staff for ESMP	PIU																	
19.	Review & Approval of Contractor's ESMP, Waste & Safety Plan	PIU																	
20.	Finalization of Engineering Designs	PIU/Engineering Design Consultant																	
21.	Environmental and Social Training	E&S Consultant																	
22.	Mobilization to site	Contractor																	
23.	Site Clearing	Contractor																	
24.	Construction Phase	Contractor																	
25.	Implementation of Mitigation	Contractor																	
26.	Supervising ESMP Implementation	PIU																	
27.	Monitoring & Reporting on ESMP Implementation	PIU/Relevant MDAs																	
28.	Environmental and Social Auditing	PIU/E&S Consultant																	

5.8 Contractual measures

Most of the mitigation measures are the obligation of the Contractor during all phases of the project. Consequently, the potential contractor will have to prepare their proposals taking into account the measures in the ESMP Tables as well as the detailed general environmental management conditions during civil works attached as Appendix 4

5.9 ESMP Disclosure

After the ESMP review and clearance by the World Bank, the following below in Table 5.8 describes the process of disclosure.

Table 5.8: Disclosure procedure

s/n	Action	Remarks
1	Disclosure on 2 national newspapers	The PIU in coordination with the FMEnv will disclose the ESMP as required by the Nigeria EIA public notice and review procedures
2	Disclosure at the Enugu State Ministry of Environment	The PIU in coordination with the FMEnv will disclose the ESMP as required by the Nigeria EIA public notice and review procedures
3	Disclosure at Nsukka LGA office	The purpose will be to inform stakeholders about the project activities; environmental and social impacts anticipated and proposed environmental and social mitigation measures.
4	Disclosure at Deputy VCs office, UNN	The purpose will be to inform stakeholders within the campus (students and staff) about the project activities; environmental and social impacts anticipated and proposed environmental and social mitigation measures.
5	Disclosure at the World Bank external website	The ESMP will be disclosed according to the World Bank 2010 Access to information Policy

CHAPTER SIX: GRIEVANCE REDRESS MECHANISM

6.1: Introduction

This section provides the procedures for receiving, hearing and resolving disputes or complaints arising from the implementation of this project. ACE-SPED recognizes the right of individuals, households, students and the community to fair hearing and justice on issues that affect them that is caused by the project. Potential sources of grievances include unethical behaviours from the contractor or contractor workers, violation of agreements, gender-based violence and sexual exploitation and non-adherence to mitigation measures contained in this ESMP.
The UNN does not have an established GRM but follows an adhoc procedure, whereby an aggrieved person, for example a student, is required to report to the dean of students affairs, who then constitutes a committee to listen to the disputing parties. In terms of contractor-client disputes it is managed through the contract arbitration clause. Although contracts in this project have arbitration clause embedded in them, setting of grievance redress committee early during the project's preparation is imperative to ensure rapid resolution of disputes and ensuring project sustainability.

6.2: Grievance Channels

The grievance redress mechanism for UNN ACE-SPED shall comprise of two channels: project level and the senate level. The project level shall constitute of the following committee membership:

- Head of University's Legal Unit- Chairman
- Dean of Students Affairs
- Head/director of gender studies
- ACE-SPED Project Manager
- Environmental safeguard officer - Secretary

6.3 Informing Parties on Levels and Channels of Grievance Uptake

Community members and contractors have been sufficiently informed during public consultation meetings held in the project area on how to channel grievances. Aggrieved person (s) will lodge their complaints to either the office of the chairman of the committee or to the dean of students Affairs. Complaint boxes shall be provided in both places. Aggrieved persons are also free to upload their complaints through email at *acesped@unn.edu.org*

Grievance redress shall be funded by the PIU so that there shall be no cost to the aggrieved/complainant for redress. Stakeholders have also been informed that it is their right to seek redress in the court of law as the last resort, if they felt dissatisfied with the judgments obtained from the grievance redress committees set up by this project.

6.4: Grievance Reporting and Redress Protocol

The procedure for reporting and addressing potential grievance arising from this project shall involve the steps described in the grievance log shown below:

Figure 6: Grievance Log showing steps for grievance redress
Source: Adopted from World Bank Policy Guidance

As shown in the grievance log, an aggrieved party will have the opportunity to lodge complaint with the GRC at the Chairman's office (Head of University Legal Unit). That will be the first channel of grievance uptake. It is expected that the matter should be addressed and determined within 14 days from date of receiving the complaint. If a complainant feels dissatisfied with the outcome of the closure of the matter by the project GRC, he/she is encouraged to report to the school Senate for redress within 14 days. The University is an establish structure and the existing structure will be sufficient to hear and determine complaints arising from the implementation of ACE-SPED, whereby a matter is not resolved at the project level. If unsatisfied with the outcome of the Senate's GRC intervention, the aggrieved party could pursue his/her right in a court of jurisdiction as the last resort. The following are the protocols for receiving and treatment of grievances:

i. Registration

The secretary of the committee will receive grievance from the complainant, register and acknowledge receipt of grievance to the aggrieved within 3 days. The registration will capture the following data: **name of the complainant, date of the grievance, category of the grievance,**

persons involved, and impacts on complainant life, proofs and witnesses. A registration form will have all these bits of information.

ii. Treatment of Grievance

This involves the verification of grievance to determine among other things whether the matter has relationship with the project activities, and whether the matter can be handled/resolved at the level where it is presented. This will determine if the matter should be referred to the next level or not. Part of the investigations may also be to assess the cost of lost or risk involved in the grievance.

iii. Closing of the Grievance or the Processing of the matter

This involves options and approach taken to resolving the case. This may involve site visit for physical inspection and determination of the claim, negotiation with the aggrieved person(s) and feed back to the parties involved.

iv. Feed back

All responses to the complainant in a grievance redress process that moves beyond a unit level must be communicated in writing and/or by verbal presentation to the complainant. This will include a follow up on the corresponding authority where cases are referred, to ascertain the status of reported cases. Feedback on outcome of each case should get to the complainant through the secretary of committee or environmental contact/safeguard person as the case may be. It is expected that reported complaints at each level will be resolved and determined within **14 days** from date of receipt of the complaint.

6.5: Financing of the Grievance Redress Mechanism and Cost of Remediation

The proponent shall be responsible for the funding of logistics for the GRC as well as the eventual compensation or remediation that aggrieved party may be entitled to. The proponent will also be responsible for the cost of the judicial process for cases that result to court for adjudication. The anticipated cost of GRM is 5% of the project mitigation cost. The implementation plan for the GRM is shown in Table 6.1.

Table 6.1: Implementation Plan for Grievance Mechanism

Steps	Process	Description	Completion Time frame	Responsible Agency/Person
1	Receipt of complaint (at Project level)	Document date of receipt, name of complainant, address/designation, nature of complaint, inform the PIU	1 day	Secretary to GRC at project level
2	Acknowledgement of grievance	By letter, email, phone	1-3 days	Environmental & Social safeguard Officer at PIU
3	Screen and Establish the Merit of the Grievance	Visit the site; listen to the complainant/community; assess the merit	5-14 days	GRC & Environmental & Social safeguard Officer& the aggrieved person or his/her representative
4	Implement and monitor a redress action	Where complaint is justified, carry out redress action	14 days or at a time specified in writing to the aggrieved Person	Project manager -PM-PIU and Environmental & Social safeguard Officer
5	Extra intervention for a dissatisfied scenario	Matter reported to the UNN Senate ● Review the redress steps and conclusions, ● provide appropriate redress	14 days of receiving status report	PM-PIU
6	Judicial adjudication	Take complaint to court of law	No fixed time	Complainant
7	Funding of grievance process	GRC logistics and training, redress compensation, court process	No fixed time	The proponent

CHAPTER SEVEN: CONSULTATION WITH STAKEHOLDERS

7.1: Introduction

This chapter gives the details of stakeholders' identification and consultations, including a summary of the concerns raised and how those concerns were addressed. Stakeholder consultations were held within the project area between the 2nd and 11th November 2022. The agenda was to share project information, elicit views and concerns about the proposed intervention from the affected residents and students within the UNN university community. Through this process, concerns and issues raised were addressed while views and inputs formed part of the mitigation/enhancement measures in this ESMP. The method used for stakeholders' data gathering was in-depth checklist, questionnaire and participatory consultation.

7.2 How Stakeholders were Categorized and Identified

The stakeholders were categorized into primary and secondary stakeholders. Primary stakeholders are those that are directly affected by the adverse impacts of the proposed works such as residents around the project site and students living in the hostels mapped for renovation and those participating or benefiting from the project. On the other hand, secondary stakeholders are those with some form of interest and influence on the project such as the UNN management, the Security Team, State Ministry of Environment and Enugu state waste management Agency (ESWAMA).

7.3 Summary of Stakeholder Consultation

The summary of the consultation is presented below:

LOCATION OF MEETING	FACULITY OF ENGINERRING BUILDING UNN,ENUGU STATE
MEETING DATE AND TIME	5TH November, 2022 10am -12.10pm
LANGUAUGE	English
PARTICIPANTS	Centre Administrators and staff members, residents of streets adjacent to the project site, UNN security agents, representatives of the management of the University executive guest house (Presidential lodge), residential students at the proposed hostel for renovation (Scholar Village), members of staff whose offices/residences are in close proximity to the project site, some beneficial ACE-SPED students

CONCERNS / COMPLAINTS	1. A concerned member of staff complained that the movement of heavy-duty vehicles may cause traffic and disturbances to academic activities. 2. The residents and students raised the issue of particulate dust during construction as well as stones flying from big truck when trucks begin to convey granite to the project site. 3. The residents raised security concerns with the likely influx of workers to the project site. 4. Concern about gender-based violence and sexual exploitation was also raised and discussed given the fact that there will be influx of migrant workers in an environment where there are many female students. 5. There was concern about where the students will be accommodated while the renovation of the hostels begins. During the construction, there maybe a case where water pipes or sewage pipes, electricity armoured cables are damaged.
HOW CONCERNS WERE ADDRESSED	1. Although the route to the project site is considered a less prone traffic area, the contractor will limit movement of heavy-duty vehicle during the activity hours of the day. In addition, there will be effective traffic management control using the traffic personnel of the contractor. The contractor will ensure that construction works are not carried out in the night in other not to escalate noise impact while adopting or using noise proof equipment as much as possible. 2. The contractors will ensure that trucks conveying granite to project sites are properly covered with tarpaulin, the contractor will ensure that the ground is properly wet regularly during construction begins as this will help to minimize particulate dust in the neighbourhood. Also, there will be metal or zinc barricade of the project site during construction. 3. ACE-SPED in collaboration with the contractor will work with the UNN security network to ensure necessary profiling of all movement (vehicle and pedestrians) into the site/school 4. The contractor staff will be sensitized on proper code of conduct and will undertake a committal agreement on that. Also, there will be a clause in the contractors contract specifying stringent penalty to breach of GBV/SE. 5. There are four hostels out of which two will be renovated. During the renovation work, the students will be moved to other hostels that are not part of the renovation exercise. The contractor will also be encouraged to expedite action on the renovation work to ensure timely completion of work .
CONCLUSION	The information shared about the project benefits and how to mitigate the potential adverse impacts were satisfactory to the stakeholders. It was hoped that the ESMP will capture all the agreements. On this note, the meeting came to a close at 12.10pm.

PICTURES FROM CONSULTATION

Pictures taken during the stakeholders consultation

CHAPTER EIGHT: CONCLUSION AND RECOMMENDATIONS

The proposed African Centre of Excellence for Sustainable Power and Energy Development (ACE-SPED) will have highly beneficial impacts on the students, host community (University of Nigeria, Nsukka), Enugu State and Nigeria at large as it will promote access to sustainable power and energy in addition to enhanced infrastructure for academic learning. Adverse environmental and social issues identified in this study will be managed through well-tailored and specific mitigation measures. Success of the ESMP will be achieved through a system of tracking of implementation and monitoring mechanism as spelt out in this report, while any observed changes during implementation should be flagged for quick and timely adjustment. Mitigation measures in this ESMP are specific to identified impact. Key highlights of the mitigation measures include:

- Priority should be given to local workers during project implementation in order to limit the number of migrant workers. This will reduce threats to community culture, health, safety and security as well as stimulate local socioeconomic activities, improve livelihood and reduce poverty.
- Water should be used for dust suppression during civil works especially those involving excavations and other dust generating activities in order to protect residents, students and members of the public from respiratory diseases and other health related challenges of dust.
- Develop Codes of Conduct for contractors with prohibitions against GBV/SEA.
- Implement GRM for effective resolution of grievances.
- The Safeguard Unit of PIU should ensure active monitoring to ensure the contractor adhere strictly to the requirements of this ESMP especially in the application of mitigation measures during project implementation.
- Ensure that trucks carrying loose and hazardous materials (soil and gravels) to the site are covered with trampoline.
- Ensure that staging areas for contractor equipment are adequately delineated and cordoned off with reflective tapes and barriers
- To enhance the positive gains of the project such as employment of local workers and human capacity building, the PIU can consider the issuance of Testimonials/Certificate to the trained local workers and/or those that worked on the project, to give them advantage in securing jobs elsewhere after the end of the ACE-SPED project.

APPENDICES

Appendix 1: Terms of Reference

1.0 Introduction

The Africa Center of Excellence for Sustainable Power and Energy Development (ACE-SPED) is based in the main campus of the University of Nigeria, Nsukka. The Centre was conceptualized to proffer sustainable solutions to some development challenges peculiar to the Sub-Saharan Africa region and it draws her staff from the University of Nigeria and operates based on already established managerial and fiduciary practices in the University. These developmental challenges faced by majority of countries in the region include lack of access to reliable electric power supply, poor diffusion of renewable and alternative energy technologies, as well as shortage of high-level skilled manpower to deploy and manage the power supply system.
The fundamental aims of ACE-SPED are; to carry out impactful educational, research, development and training activities in five major thematic research areas, namely (i) Electric power systems development; (ii) Renewable energy, waste-to-energy and energy conservation; (iii) Energy resources assessment and forecasting; (iv) Sustainable energy materials and (v) Energy Policy, Regulation and Management. Using a student-centered, interactive and engaging educational curriculum cum robust Research & Development themes with strong practice orientation, ACE-SPED plans to equip its Master's and PhD degree graduates to deliver great impact on sustainable power and energy development in the Sub-Saharan Africa region in the medium to long terms. These justify the urgency for an enabling infrastructure for administration, research, teaching, learning and hostel accommodation.

The ACE-SPED UNN building project involves the construction of the main building and the upgrade and rehabilitation of the existing hostels to be utilized for ACE-SPED UNN activities. This project will have significant environmental and social impacts. To ensure all impacts are identified and mitigation measures as well as responsibilities for implementing these measures are in place an Environmental and Social Management Plan (ESMP) is required. The ESMP will also include recommendations on good work practices considering the COVID-19 situation which, requires physical/social distancing, good hygiene practices and consultations where required. The ESMP will ensure the ACE SPED is in compliance with the World Bank Safeguard Policies particularly OP 4.01 Environmental Assessment and OP 4.11 Physical and Cultural Resources triggered for the ACE II Project. It will also ensure the project complies with the Nigeria EIA Act CAP. E12 L.F.N. 2004.The ESMP will provide an overview of the environmental and social baseline conditions of the proposed project, summarise the potential impacts associated with the proposed construction works and set out the management measures including implementation and responsibilities required to mitigate any potential risks and impacts associated with the activities of the factory.

In addition, the ESMP will be utilised by the contractor, to be commissioned by ACE SPED for the sub-project, and will form the basis of site-specific management plans that will be prepared by the contractors as part of their construction methodology prior to works commencing also taking into cognisance the COVID 19 pandemic and the different measures used in preventing its spread.
Thus, ACE SPED intends to engage a qualified consultant to carry out this ESMP.

DESCRIPTION OF THE PROPOSED SUB-PROJECT ACTIVITIES AND RATIONALE FOR THE STUDY

The proposed sub-project will contain the following features:

- Two Lecture Theatres and two Halls for lecture delivery.

- Five Laboratories on the ground floor
- One Library for postgraduate students
- Administrative and Professorial Offices for staff
- One large Conference Room and one conference room in the Director's Office
- Workshops
- Rehabilitation of existing hostels

It is estimated that the project will cost USD 1.35 million. The activities of the proposed project is anticipated to have environmental and social impacts. Therefore, in compliance with the requirements of the Nigerian EIA laws and the World Bank Environmental and Social Safeguard Policies OP 4.01 Environmental Assessment and 4.11 Physical Cultural Resources both triggered on this Project, the ACE-SPED UNN, is proposing to award a contract to prepare an Environmental and Social Management Plan (ESMP) which will identify the environmental and social impacts associated with the project and also proffer mitigation measures to address these impacts. . The proposed ESMP is to provide an overview of the environmental and social baseline conditions of the proposed sub-projects, summarise the potential impacts associated with the proposed construction and furnishing works and set out the management measures required to mitigate any potential impacts in a series of sector specific Environmental and Social Management Plans (ESMPs).

These ESMPs are to be utilised by the contractors, to be commissioned by ACE-SPED UNN for the project, and will form the basis of site-specific management plans that will be prepared by the contractors as part of their construction programme of work and methodology prior to the project commencement date.

As the proponent for the ACE-SPED UNN Construction project, it is ACE-SPED UNN's objective to avoid, where feasible, unacceptable adverse environmental, social and/or economic impacts. In the circumstance that an impact cannot be avoided, ACE-SPED UNN and the Project Manager-PM (who will be responsible for the management of the procurement of the ACE-SPED UNN project) are committed to the implementation of appropriate mitigation measures. For clarity in the management structure however, the PM will consult ACE-SPED UNN on matters relating to environmental health and safety performance. The PM will however have overall responsibility for planning, implementation, monitoring and enforcement of activities associated with the ESMP and environmental and health and safety performance.

The Project site location is shown in Fig. 1 below.

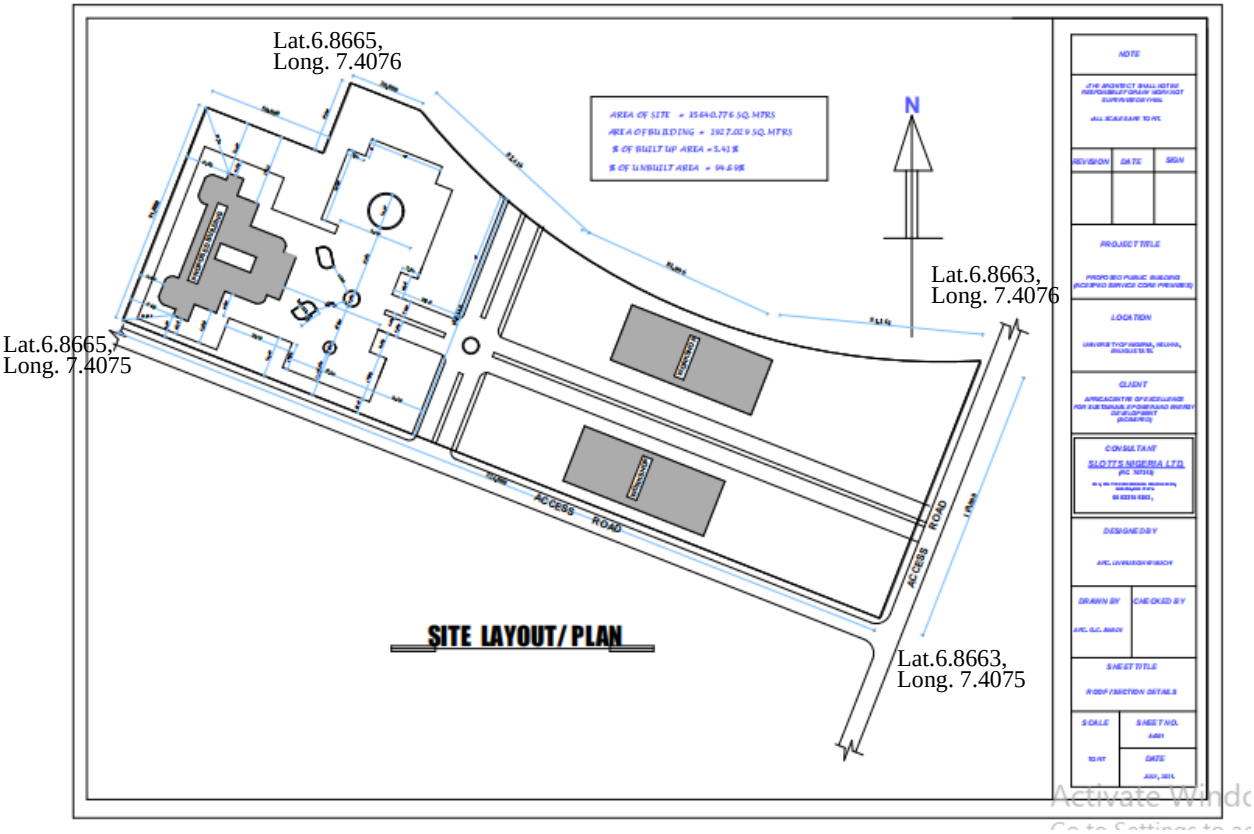


Fig. 1: Site map of ACE-SPED Project

3.0 Terms of Reference for the ESMP are to:

Describe the measures required to implement the construction of the ACE-SPED UNN building project and the upgrade and rehabilitation of the existing hostels to be utilized for ACE-SPED UNN activities and sub-projects, related management and mitigation commitments;

Describe specific additional measures required to implement contract related international best practices, and approval conditions stipulated by the World Bank Safeguard Standards, Nigeria's Federal Ministry of Environment and NESREA requirements;

Identify the roles and responsibilities of the environmental and social management stakeholders of the project; and

Communicate the environmental and social expectations and requirements throughout the project tenure.

While all contractors and subcontractors shall comply with, and apply the ESMP requirements as applicable to the tasks they are employed to undertake, some of the measures and procedures outlined in this proposal are commitments made by the PM of ACE-SPED UNN, and therefore remain responsible for their implementation. It should however be recognised that practical implementation of many of the measures may rest with contractors and subcontractors and consequently, ACE-SPED UNN will require the implementation of a robust review/audit programme, as may be described in the proposed ESMP, to measure and ensure that they are executed on their behalf properly.

4.0. OBJECTIVES OF THE CONSULTANCY

The ESMP will achieve the following objectives:

- Potentially screen out environmentally unsound activities

- Proposes modified designs to reduce environmental and social impacts
- Identifies feasible alternatives
- Predicts significant adverse impacts
- Identifies mitigation measures to reduce, offset, or eliminate adverse impacts
- Engages and informs potentially affected communities and individuals
- Influences decision-making and the development of terms and conditions
- - Provide a detailed description of the main phases of the works (preparation, execution/construction, operation) and key activities of the works to be executed;
- - Conduct public consultations with all the stakeholders of the Project in order to collect their suggestions that will be integrated in the ESMP;
- - Carry out an analysis of the initial environmental status of the sites concerned and describe the biophysical, socio-economic and cultural characteristics of the environment in which the sub-project activities will take place,
- - Highlight the major constraints that need to be taken into account at the time of site preparation, construction and during the construction and installation of equipment and during the operation phase;
- - Identify the environmental and social components that may be impacted by the implementation of the works
- - Check to ensure that the design and construction methods and the various components of the works are consistent with the ecological, social, economic and cultural realities of the project areas and, if necessary, propose any necessary readjustments;
- - Analyse the legal and regulatory framework for environmental management in relation to national legislation and World Bank safeguard policies;
- - Identify and analyse, by implementation phase, the potential social and environmental impacts, both positive and negative (including impacts on physical cultural resources) that may result from the realisation of the rehabilitation, renovation or construction works of the infrastructures
- - Assess the significance of these impacts during the project phases, compared to the no-project scenario;
- - Propose measures to optimize the positive impacts during the different phases of the works and technically viable and economically feasible mitigation measures to avoid, minimize, mitigate or compensate for the negative environmental and social impacts so as to bring them to acceptable levels; on the environment and the human environment;
- - Present the costs of implementing the proposed mitigation measures;
- - Assess the need for solid and liquid waste collection, disposal and infrastructure management, and make recommendations;
- - Propose a mechanism for managing complaints that may arise in the course of the works.

4.0 SCOPE OF WORK:

The core tasks for the ESMPs shall include:

- Assess the potential environmental and social impacts related to the construction and rehabilitation activities and recommend adequate mitigation measures, including costs estimation.
- Identify responsibilities and actors for the implementation of proposed mitigation measures.
- Assess the resources and competencies available to implement the proposed mitigation measures, and suggest recommendation in terms of training and capacity building, and estimate their costs.

- Develop an Environmental and Social Management Plan (ESMP) for the work. The ESMP should underline
  (i) the potential environmental and social impacts resulting from construction and rehabilitation activities
  (ii) the proposed mitigation strategies
  (iii) the institutional responsibilities for implementing of the ESMP
  (iv) the key performance indicators for monitoring and control
  (v) the institutional responsibilities for monitoring and implementation of mitigation measures;
  (vi) the costs of implementing the ESMP; and
  (vii) the calendar of implementation.
- Consultations: The ESMP results and the proposed mitigation measures will be discussed with relevant stakeholders mainly involved by the rehabilitation activities. Recommendations from this consultation will be included in the final ESMP report.
- Preparing the ESMP per the generic contents below:

The objective of the study is to prepare an Environmental and Social Management Plan (ESMP) for the proposed ACE-SPED UNN project.

The ESMP should consist of a well-documented set of mitigation, monitoring, and institutional actions to be taken before and during rehabilitation to eliminate adverse environmental and social impacts, offset them, or reduce them to acceptable levels. It should also include the measures needed to implement these actions, addressing the adequacy of the monitoring and institutional arrangements at the permanent office site.

Methodological approach to conduct the study

The consultant is requested to describe precisely and clearly each of the methods and tools he will use for both data collection and data processing. He will examine the interactions between the project's nuisance emitters and the environmental receptors subject to the corresponding interference, while excluding those aspects which have little or no relevance to the environmental impacts of the proposed action. It will identify those elements of the biophysical and social environment which may be affected by the project and for which there is public and/or professional concern. It will identify all the potential environmental impacts of the project and assess them using an appropriate methodology to rank them in order of importance. Only significant impacts will be examined in depth. The Consultant will then propose realistic and feasible mitigation or enhancement measures and a monitoring program for these.
A particular attention will be paid to sensitizing the population in the project area and the drivers of construction machines and vehicles on environmental protection and safety aspects. The Consultant will assess the risks associated with the project and propose measures to be taken in case of emergency. The Consultant will propose responses to the environmental feasibility of the project.
The Consultant is advised to use the Participatory Research Method (PRM) for the collection of environmental information.

The study will also propose, among others, the following
- a waste management plan for the project's activities
- a gender and vulnerability assessment plan
- a Grievance Redress Mechanism.

The study will be conducted in compliance with the environmental impact assessment procedures developed by the country. It should also take into account the World Bank's OPs triggered by the project.

The methodology adopted by the consultant will be rigorous and will involve a study of the initial state, the identification of potential impacts, the evaluation of relevant impacts and the identification of insertion measures.
Therefore, the consultant will cover by all means (vehicles and/or feet) the whole area of influence of the project previously well defined in agreement with the Promoter

Legal and Institutional Framework

This study will be guided by the World Bank operational safeguard policies, and the Enugu State Government and Federal Government of Nigeria policies and regulatory framework. The project will comply with the Nigerian Environmental Impact Assessment Act CAP. E12 L.F.N. 2004, The National Policy on the Environment, The Land Use Act (Cap 202, 1990), now Cap L5 Laws of the Federation of Nigeria 2004, Harmful Waste (Special Criminal Provisions) ACT CAP H1, LFN 2004, Labour Act - CAP. L1 L.F.N. 2004, Workers' Compensation Act, 1987 and EIA Procedural Guidelines.

The institutional responsibilities for the environmental components of the project will be shared by the three tiers of government: federal, state and local governments. At the federal level, the institutions that will be responsible for this project include the National Council on Environment, the Federal Ministry of Environment and its agency, the National Environmental Standards and Regulations Enforcement Agency (NESREA) which will play a key role of ensuring that the environmental and social requirements of the project are fully adhered to. At the state level, the Enugu State Ministry of Environment and Enugu State Environmental Planning and Protection Agency will provide the institutional framework for the project.

This project is classified under the World Bank safeguards policy as category B, as its adverse effects on the population or areas of environmental importance are limited, site-specific, and likely reversible, and the mitigation measures can be easily designed and implemented. This project triggers World Bank safeguards Operational Policy (OP) and Bank Procedure (BP), OP/BP 4.01 (Environmental Assessment) that covers impacts on the environment, human health and safety of the project. In addition, the Bank's Environmental, Health and Safety Guidelines, the 2010 Access to Information Policy, and the Guidelines on Labour Influx will all be applied.

TABLE 1: The Roles and Responsibilities of the Stakeholders in the Implementation, Monitoring and Review of the ESMP

S/N	Stakeholder	Roles and Responsibilities
1	ACE-SPED	● Liaise closely with Enugu State Ministry of Environment in preparing a coordinated response on the environmental and social aspects of project development respectively as required; ● Safeguards due diligence ● In case of any violations on arising works, PIU will request contractors to amend and correct the violation. ● Receive and supervise the environmental report from the Independent Environmental Consultant (IEC), ● PIU's Safeguard Specialist will be in charge of review of environmental report and recommend further actions. ● Cooperate with WB to periodically supervise contractors' activities.

		● Weekly meetings will be held between Contractors, PIU, and consultants. In addition, PIU is to have assigned personnel for regular inspection.
2	Enugu State Ministry of Environment	● Environmental compliance Coordinator at the State level ● Lead role - provision of advice on screening, scoping, review of draft ESMP report (in liaison with Federal Ministry of Environment) ● Site assessment and monitoring of ESMP implementation
3	Federal Ministry of Environment	● Lead role - provision of advice on screening, scoping, review of draft ESMP report (in liaison with State Ministry of Environment), ● Receiving comments from stakeholders, public hearing/ review of the project proposals, and convening a technical decision-making panel arising from the public disclosures, ● Project categorization for EA, Applicable standards, Environmental and social liability investigations, Monitoring and evaluation process and criteria
4	Other MDAs	● Other MDAs come in as at when relevant areas or resources under their jurisdiction or management are likely to be affected by or implicated by projects. ● They participate in the EA processes and in project decision-making that helps prevent or minimize environmental and social impacts and to mitigate them. These institutions may also be required, to issue a consent or approval for an aspect of a project; allow an area to be included in a project; or allow impact to a certain extent or impose restrictions or conditions, monitoring responsibility or supervisory oversight
5	World Bank (WB)	● Overall supervision and provision of technical support and guidance. ● Recommend additional measures for strengthening the management framework and implementation performance; ● Supervising the application and recommendations of sub- project ESMPs.
6	Independent Environmental Consultant	● Make reference and prepare site specific plans in accordance with ESMP framework approved by WB. The respective contractors will be required to formally commit to requirements of the site-specific Management Plans. ● Establish environmental procedure and notify and obtain formal commitment from contractors. ● Observe directly the performance of the environmental works, ● report mitigation measures of the environmental impact result from contractor's activities; ● Review weekly environmental reports prepared by the Principal Contractor;

		● Undertake assessments and audits and issue environmental supervision reports on a weekly basis ● Report to PIU / World Bank as may be necessary the performance of the contractors and recommend countermeasures if any.
7	Safeguards Unit	**Environmental Aspects** ● Collate environmental baseline data on relevant environmental characteristics of the selected project sites; ● Analyze potential community/individual sub-projects and their environmental impacts; ● Ensure that project activities are implemented in accordance to best practices and guidelines set out in the site specifics; ● Identify and liaise with all stakeholders involved in environment related issues in the project; and ● Be responsible for the overall monitoring of mitigation measures and the impacts of the project during implementation. **Social Aspects** ● Develop, coordinate and ensures the implementation of the social aspects of the ESMP; ● Implement the grievance mechanism on the project ● Identify and liaise with all stakeholders involved in social related issues in the project; ● Conduct impact evaluation and beneficiaries' assessment; and ● Establish partnerships and liaise with organizations, Community Based Organizations (CBOs) and Civil Society Organizations (CSOs).
8	Works Contractor	● After receiving and committing to the environmental procedures and Management Plans, ● Contractors must fully carry out the measures of the environmental protection, health and safety procedure as indicated; ● Any changes related to Environment, health and safety procedure must be informed to PIU for approval ● In case of any violations on arising works either detected by Environmental consultant, Safeguard Specialists, or new arising works proposed by contractors, they must be reported to PIU for further actions. ● If contractors decided to not follow instruction from PIU, construction activities will be halted until necessary actions are taken.

		• Contractors must assign at site, personnel in charge of Environment, health and safety procedure. • Compliance to BOQ specification in procurement of material and construction
9	Site Engineer	• Provide oversight function during construction and decommissioning
10	NGOs and CBOs	• Assisting in their respective ways to ensure effective response actions, • Conducting scientific research alongside government groups to evolve and devise sustainable environmental strategies and rehabilitation techniques, • Organizing, coordinating and ensuring safe use of volunteers in a response action, and actually identifying where these volunteers can best render services effectively & providing wide support assistance helpful in management planning, institutional/governance issues and other livelihood related matter, • Project impacts and mitigation measure, Awareness campaigns

5.0 ESMP Structures

The ESMP Report shall be presented in a concise format and should not be more **than 20 pages** containing all studies, processes, analyses, tests and recommendations for the proposed intervention. The report shall focus on the findings, conclusions and any recommended actions, supported by summaries of the data collected and citations for any references used. The ESMP report will include the following topics.

Preliminary pages

- Cover page
- Table of contents
- List of acronyms and their definitions
- Executive Summary

Chapter 1: Introduction

- Background
- Rationale for ESMP
- Scope of the ESMP
- Discussion of the World Bank safeguard policies triggered by the Project and the proposed sub-project activity
- Applicable Environmental and Social Laws and Regulations

Chapter 2: Project Description

- Description of the proposed construction and equipping the ACE Design Studio Laboratories activities
- Project Activities and Schedules
- Relevant Maps and engineering designs for proposed construction activities.

Chapter 3: Baseline Description of Project Area

- Introduction
- Overview of the Project Area including relevant maps and engineering designs for proposed construction activities.
- Socioeconomic Environment of the Project Area
- Biophysical Environment of the Project Area
- Environmental and Social Sensitivities in the project areas (including map of the hubs showing the

locations of all sensitivities)

- Gender and Gender Based Violence Statistic including state information as well as those related to the project
- Implication of baseline on the Proposed project activities
- Description of the area of influence and environmental and social baseline conditions

Chapter 4: Assessment of Potential Adverse Environmental and Social Impacts.

- Methods and techniques used in assessing and analyzing the environmental and social impacts of the proposed construction.
- Discussion of the potentially significant adverse environmental and social impacts of the proposed construction.

Chapter 5: Environmental and Social Management Plan (ESMP), including:

- generality
- Identification of positive impacts
- Identification of negative impacts
- Environmental and Social Management Table (ESMP table) highlighting Environmental and Social Management Table (ESMP table) highlighting Activities, identified adverse impacts, mitigation Measures and corresponding Indicator(s), Mode of Measurement, corresponding Cost of Mitigation, monitoring indicators, Frequency, Monitoring Cost, as well as responsibilities for implementing these measures
- Roles and Responsibilities
- Capacity Building
- Institutional responsibilities for monitoring and implementation of mitigation;
- Monitoring and Reporting
- Implementation Schedule
- Contractual Measures
- Measures for Non-Compliance with the ESMP
- Cost Estimates for ESMP Implementation
- Monitoring indicators;

Chapter 6: Grievance Redress Mechanism

- Introduction
- Expectation When Grievances Arise
- Structure and Protocols for Reporting and Managing Grievances
- Protocol for reporting and managing grievance for this sub project.

Chapter 7: Consultation with Stakeholders

- This chapter shall summarize the actions undertaken to consult the groups affected by the construction. The detailed record of the consultation meetings shall be presented in annex to the ESM

Chapter 7: Conclusion and Recommendations

The Annexes below must consider COVID-19 requirement. See Annex K for WHO, World Bank and ILO guidance notes on construction, labour, waste management, etc. provisions.

Annex 1: Terms of Reference
Annex 2: Sample of Questionnaire for socio-economics
Annex 3: List of participants in consultations and summaries of meetings
Annex 4: General Environmental and Social Management Conditions for Construction Contracts
Annex 5: Project Occupational Health and Safety (OHS) Plan
Annex 6: Company Code of Conduct on Preventing Gender Based Violence and Violence Against Children
Annex 7: Manager's Code of Conduct on Preventing Gender Based Violence and Violence against Children
Annex 8: Individual Code of Conduct on Preventing Gender Based Violence and Violence against Children
Annex 9: Waste Management Plan
Annex 10: Traffic Management Plan
Annex 11: Workers Campsite Management Plan
Annex 12: Safeguard guidance on Covid-19 consideration in construction/civil works projects
Annex 13: List of Persons Met
Annex 14: References
Annex 15: Photos
Annex 16: Traffic Management Plan
Annex 17: Chance Find Procedure

Appendix 2A: Checklist/Questionnaire for Impact Identification

PART A: BRIEF DESCRIPTION OF THE ENVIRONMENTAL SITUATION AND IDENTIFICATION OF ENVIRONMENTAL AND SOCIAL IMPACTS

List the sub-projects to be implemented under this project:

1……
2……
3……
4……
5…….

Describe the sub-projects in terms of: location, green or brown project, size of land, (include a sketch map and take GPS of each location)

Describe the land formation, topography, and vegetation in or adjacent to the project area.

Estimate and indicate where vegetation might need to be cleared (if any)

PART B: CHECKLIST

S/N	Description	Yes/No	Remark
1	Any existing state of soil erosion within the project area?		
2	Has there been any case of flooding in the project site within the last 5 years?		
3	Is there a river/surface water body within 1km from the project site?		
4	If yes to Q 3, state (in the remark column) the various uses to which the water body is used. Also state the name of the water body		
5	Does UNN ACE-SPED has an Environmental Officer?		
6	If yes to Q 5, When last (year) did the officer receive training on environmental management?		
7	If no, do you have a plan to hire an environmental expert or officer?		
8	Sub-projects such as laboratory and hotels involve emission and disposal of waste water. Do you have a process or a plan for waste water treatment/ management?		
9	Are you going to use direct labour or a contractor to implement the rehabilitation and construction civil works?		
10	PPEs such as face mask, hand gloves, helmet, safety boots will be required for personnel undertaking rehabilitation work, who will be responsible for supply of PPEs (ACE-		

	SPED or contractor?		
11	Is there a budget for PPEs?		
12	Grievances may arise during construction between client and contractor, or between construction employees and contractor. Is there any existing mechanism in place at UNN for managing conflicts/grievances?		
13	If yes to Q12, briefly describe the process of grievance/conflict redress in place		
14	Influx of workers and visitors during construction and operation stages of the project, may expose the students to interaction with visitors/workers, which may result in gender based violence and sex exploitation and assault. (GBV/SEA). In the unexpected event of GBV/SEA, is there a GBV referral facility and or trained personnel for rehabilitation of victims? If yes, where is the place?		
15	Who will benefit from ACE-SPED projects in UNN?		
16	What is the population of students of the ACE-SPED?		
17	State the gender ratio of the population		

PART C: TAKE PICTURES AND NAMES OF PEOPLE CONTACTED

1. Picture of before project situation of the facilities to be rehabilitated
2. Picture with those contacted at ACE-SPED
3. Attendance/Contact (Name, designation and signature) of those contacted.

Appendix 2B: Risk Assessment Matrix Tool for Impact Assessment

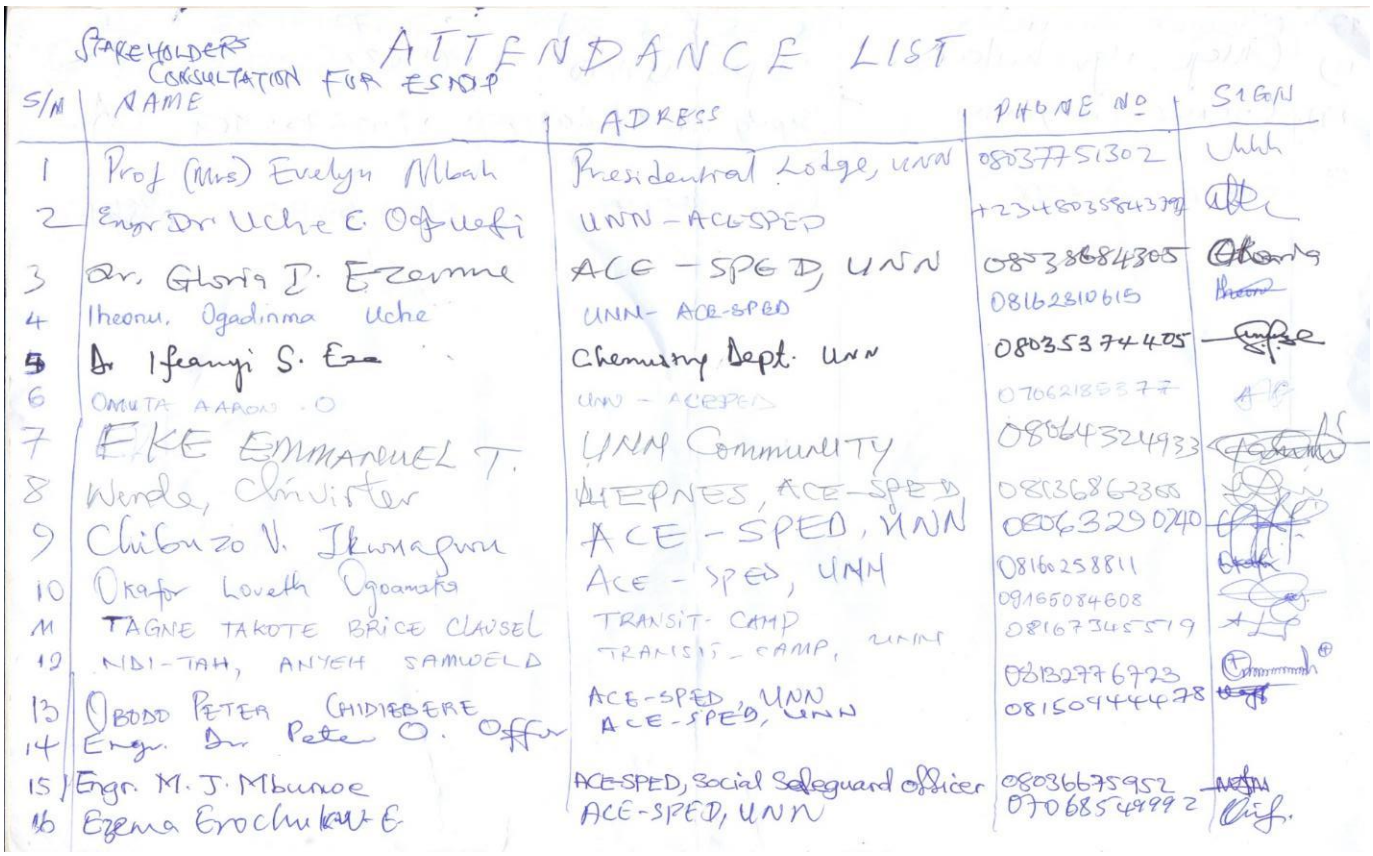
RISK ASSESSMENT MATRIX

	Probability					
Consequences		A	B	C	D	E
	1	H	H	H	H	M
	2	H	H	H	M	M
	3	H	H	M	M	L
	4	M	M	M	L	L
	5	M	L	L	L	L

Explanatory notes on the selection of the consequences and probability for each issue are presented in the Risk Matrix Explanation below:

	Probability		consequences		
A	Almost certain	Expected to occur (quite common)	1	Major	▪ Major environmental harm e.g. major pollution incident causing significant damage or potential to health or the environment ▪ Fines and prosecution likely
B	Likely	Will probably occur (has happened)	2	Significant	▪ Long term or serious environmental damage ▪ Numerous complaints received ▪ Potential for prosecution ▪ Loss of reputation
C	Possible	Might occur at sometime	3	Moderate	▪ Moderate environmental impact ▪ Will cause complaints ▪ Possible fine
D	Unlikely	Could occur at some time although unlikely	4	Minor	▪ Minimal environmental harm ▪ Potential for complaints ▪ Fine unlikely
E	Rare	Might occur at some time in exceptional circumstances	5	Insignificant	▪ Little or no environmental harm ▪ Little potential for fines or complaints

APPENDIX 3: Public Consultation Attendance List



STAKEHOLDERS CONSULTATION FOR ESMP

ATTENDANCE LIST

S/N	NAME	ADRESS	PHONE NO	SIGN
1	Prof (Mrs) Evelyn Mbah	Presidential Lodge, UNN	08037751302	
2	Engr Dr Uche E. Ofuefi	UNN - ACE-SPED	+2348035843797	
3	Dr. Gloria I. Ezeume	ACE - SPED, UNN	08038684305	
4	Iheonu. Ogadinma Uche	UNN- ACE-SPED	08162810615	
5	Dr Ifeanyi S. Eze	Chemistry Dept. UNN	08035374405	
6	OMUTA AARON .O	UNN - ACE-SPED	07062185377	
7	EKE EMMANUEL T.	UNN Community	08064324933	
8	Wende, Christer	MEPNES, ACE-SPED	08136862365	
9	Chibuzo V. Ikwuagwu	ACE - SPED, UNN	08063290740	
10	Okafor Loveth Ogoamaka	ACE - SPED, UNN	08160258811	
11	TAGNE TAKOTE BRICE CLAUSEL	TRANSIT- CAMP	08165084608	
12	NDI-TAH, ANYEH SAMUELA	TRANSIT - CAMP, UNN	08167345519	
13	OBODO PETER CHIDIEBERE	ACE-SPED, UNN	08132776723	
14	Engr. Dr Peter O. Offor	ACE-SPED, UNN	08150444478	
15	Engr. M. J. Mbunoe	ACE-SPED, Social Safeguard officer	08036675952	
16	Ezema Erochukwu E.	ACE-SPED, UNN	07068549992	

Appendix 4: General environmental management conditions for construction contracts

General

1. In addition to these general conditions, the Contractor shall comply with any specific Environmental Management Plan (EMP) or Environmental and Social Management Plan (ESMP) for the works he is responsible for. The Contractor shall inform himself about such an EMP, and prepare his work strategy and plan to fully take into account relevant provisions of that EMP. If the Contractor fails to implement the approved EMP after written instruction by the Supervising Engineer (SE) to fulfil his obligation within the requested time, the Owner reserves the right to arrange through the SE for execution of the missing action by a third party on account of the Contractor.

2. Notwithstanding the Contractor's obligation under the above clause, the Contractor shall implement all measures necessary to avoid undesirable adverse environmental and social impacts wherever possible, restore work sites to acceptable standards, and abide by any environmental performance requirements specified in an EMP. In general these measures shall include but not be limited to:

(a) Minimize the effect of dust on the surrounding environment resulting from earth mixing sites, asphalt mixing sites, dispersing coal ashes, vibrating equipment, temporary access roads, etc. to ensure safety, health and the protection of workers and communities living in the vicinity dust producing activities.

(b) Ensure that noise levels emanating from machinery, vehicles and noisy construction activities (e.g. excavation, blasting) are kept at a minimum for the safety, health and protection of workers within the vicinity of high noise levels and nearby communities.

(c) Ensure that existing water flow regimes in rivers, streams and other natural or irrigation channels is maintained and/or re-established where they are disrupted due to works being carried out.

(d) Prevent bitumen, oils, lubricants and wastewater used or produced during the execution of works from entering into rivers, streams, irrigation channels and other natural water bodies/reservoirs, and also ensure that stagnant water in uncovered borrow pits is treated in the best way to avoid creating possible breeding grounds for mosquitoes.

(e) Prevent and minimize the impacts of quarrying, earth borrowing, piling and building of temporary construction camps and access roads on the biophysical environment including protected areas and arable lands; local communities and their settlements. In as much as possible restore/rehabilitate all sites to acceptable standards.

(f) Upon discovery of ancient heritage, relics or anything that might or believed to be of archaeological or historical importance during the execution of works, immediately report such findings to the SE so that the appropriate authorities may be expeditiously contacted for fulfilment of the measures aimed at protecting such historical or archaeological resources.

(g) Discourage construction workers from engaging in the exploitation of natural resources such as hunting, fishing, collection of forest products or any other activity that might have a negative impact on the social and economic welfare of the local communities.

(h) Implement soil erosion control measures in order to avoid surface run off and prevents siltation, etc.(i) Ensure that garbage, sanitation and drinking water facilities are provided in construction workers camps.

(j) Ensure that, in as much as possible, local materials are used to avoid importation of foreign material and long distance transportation.

(k) Ensure public safety and meet traffic safety requirements for the operation of work to avoid accidents.

3. The Contractor shall indicate the period within which he/she shall maintain status on site after completion of civil works to ensure that significant adverse impacts arising from such works have been appropriately addressed.

4. The Contractor shall adhere to the proposed activity implementation schedule and the monitoring plan / strategy to ensure effective feedback of monitoring information to project management so that impact management can be implemented properly, and if necessary, adapt to changing and unforeseen conditions.

5. Besides the regular inspection of the sites by the SE for adherence to the contract conditions and specifications, the Owner may appoint an Inspector to oversee the compliance with these environmental conditions and any proposed mitigation measures. State environmental authorities may carry out similar inspection duties. In all cases, as directed by the SE, the Contractor shall comply with directives from such

inspectors to implement measures required to ensure the adequacy rehabilitation measures carried out on the bio-physical environment and compensation for socio-economic disruption resulting from implementation of any works.

Worksite/Campsite Waste Management

6. All vessels (drums, containers, bags, etc.) containing oil/fuel/surfacing materials and other hazardous chemicals shall be bunded in order to contain spillage. All waste containers, litter and any other waste generated during the construction shall be collected and disposed of at designated disposal sites in line with applicable government waste management regulations.

7. All drainage and effluent from storage areas, workshops and camp sites shall be captured and treated before being discharged into the drainage system in line with applicable government water pollution control regulations.

8. Used oil from maintenance shall be collected and disposed of appropriately at designated sites or be re-used or sold for re-use locally.

9. Entry of runoff to the site shall be restricted by constructing diversion channels or holding structures such as banks, drains, dams, etc. to reduce the potential of soil erosion and water pollution.

10. Construction waste shall not be left in stockpiles along the road but removed and reused or disposed of on a daily basis.

11. If disposal sites for clean spoil are necessary, they shall be located in areas, approved by the SE, of low land use value and where they will not result in material being easily washed into drainage channels. Whenever possible, spoil materials should be placed in low-lying areas and should be compacted and planted with species indigenous to the locality.

Material Excavation and Deposit

12. The Contractor shall obtain appropriate licenses/permits from relevant authorities to operate quarries or borrow areas.

13. The location of quarries and borrow areas shall be subject to approval by relevant local and national authorities, including traditional authorities if the land on which the quarry or borrow areas fall in traditional land.

14. New extraction sites:

a) Shall not be located in the vicinity of settlement areas, cultural sites, wetlands or any other valued ecosystem component, or on on high or steep ground or in areas of high scenic value and shall not be located less than 1km from such areas.

b) Shall not be located adjacent to stream channels wherever possible to avoid siltation of river channels. Where they are located near water sources, borrow pits and perimeter drains shall surround quarry sites.

c) Shall not be located in archaeological areas. Excavations in the vicinity of such areas shall proceed with great care and shall be done in the presence of government authorities having a mandate for their protection.

d) Shall not be located in forest reserves. However, where there are no other alternatives, permission shall be obtained from the appropriate authorities and an environmental impact study shall be conducted.

e) Shall be easily rehabilitated. Areas with minimal vegetation cover such as flat and bare ground, or areas covered with grass only or covered with shrubs less than 1.5m in height, are preferred.

f) Shall have clearly demarcated and marked boundaries to minimize vegetation clearing.

15. Vegetation clearing shall be restricted to the area required for safe operation of construction work. Vegetation clearing shall not be done more than two months in advance of operations.

16. Stockpile areas shall be located in areas where trees can act as buffers to prevent dust pollution. Perimeter drains shall be built around stockpile areas. Sediment and other pollutant traps shall be located at drainage exits from workings.

17. The Contractor shall deposit any excess material in accordance with the principles of the general conditions, and any applicable EMP, in areas approved by local authorities and/or the SE.

18. Areas for depositing hazardous materials such as contaminated liquid and solid materials shall be approved by the SE and appropriate local and/or national authorities before the commencement of work. Use of existing, approved sites shall be preferred over the establishment of new sites.

Rehabilitation and Soil Erosion Prevention

19. To the extent practicable, the Contractor shall rehabilitate the site progressively so that the rate of rehabilitation is similar to the rate of construction.
20. Always remove and retain topsoil for subsequent rehabilitation. Soils shall not be stripped when they are wet as this can lead to soil compaction and loss of structure.
21. Topsoil shall not be stored in large heaps. Low mounds of no more than 1 to 2m high are recommended.
22. Re-vegetate stockpiles to protect the soil from erosion, discourage weeds and maintain an active population of beneficial soil microbes.
23. Locate stockpiles where they will not be disturbed by future construction activities.
24. To the extent practicable, reinstate natural drainage patterns where they have been altered or impaired.
25. Remove toxic materials and dispose of them in designated sites. Backfill excavated areas with soils or overburden that is free of foreign material that could pollute groundwater and soil.
26. Identify potentially toxic overburden and screen with suitable material to prevent mobilization of toxins.
27. Ensure reshaped land is formed so as to be inherently stable, adequately drained and suitable for the desired long-term land use, and allow natural regeneration of vegetation.
28. Minimize the long-term visual impact by creating landforms that are compatible with the adjacent landscape.
29. Minimize erosion by wind and water both during and after the process of reinstatement.
30. Compacted surfaces shall be deep ripped to relieve compaction unless subsurface conditions dictate otherwise.
31. Revegetate with plant species that will control erosion, provide vegetative diversity and, through succession, contribute to a resilient ecosystem. The choice of plant species for rehabilitation shall be done in consultation with local research institutions, forest department and the local people.

Water Resources Management

32. The Contractor shall at all costs avoid conflicting with water demands of local communities.
33. Abstraction of both surface and underground water shall only be done with the consultation of the local community and after obtaining a permit from the relevant Water Authority.
34. Abstraction of water from wetlands shall be avoided. Where necessary, authority has to be obtained from relevant authorities.
35. Temporary damming of streams and rivers shall be done in such a way avoids disrupting water supplies to communities downstream and maintains the ecological balance of the river system.
36. No construction water containing spoils or site effluent, especially cement and oil, shall be allowed to flow into natural water drainage courses.
37. Wash water from washing out of equipment shall not be discharged into water courses or road drains.
38. Site spoils and temporary stockpiles shall be located away from the drainage system, and surface run off shall be directed away from stockpiles to prevent erosion.

Traffic Management

39. Location of access roads/detours shall be done in consultation with the local community especially in important or sensitive environments. Access roads shall not traverse wetland areas.
40. Upon the completion of civil works, all access roads shall be ripped and rehabilitated.
41. Access roads shall be sprinkled with water at least five times a day in settled areas, and three times in unsettled areas, to suppress dust emissions.

Blasting

42. Blasting activities shall not take place less than 2km from settlement areas, cultural sites, or wetlands without the permission of the SE.
43. Blasting activities shall be done during working hours, and local communities shall be consulted on the proposed blasting times.
44. Noise levels reaching the communities from blasting activities shall not exceed 90 decibels.
Disposal of Unusable Elements
45. Unusable materials and construction elements such as electro-mechanical equipment, pipes, accessories and demolished structures will be disposed of in a manner approved by the SE. The Contractor has to agree with the SE which elements are to be surrendered to the Client's premises, which will be recycled or reused, and which will be disposed of at approved landfill sites.
46. As far as possible, abandoned pipelines shall remain in place. Where for any reason no alternative alignment for the new pipeline is possible, the old pipes shall be safely removed and stored at a safe place to be agreed upon with the SE and the local authorities concerned.
47. AC-pipes as well as broken parts thereof have to be treated as hazardous material and disposed of as specified above.
48. Unsuitable and demolished elements shall be dismantled to a size fitting on ordinary trucks for transport.

Health and Safety
49. In advance of the construction work, the Contractor shall mount an awareness and hygiene campaign. Workers and local residents shall be sensitized on health risks particularly of AIDS.
50. Adequate road signs to warn pedestrians and motorists of construction activities, diversions, etc. shall be provided at appropriate points.
51. Construction vehicles shall not exceed maximum speed limit of 40km per hour.

Repair of Private Property
52. Should the Contractor, deliberately or accidentally, damage private property, he shall repair the property to the owner's satisfaction and at his own cost. For each repair, the Contractor shall obtain from the owner a certificate that the damage has been made good satisfactorily in order to indemnify the Client from subsequent claims.
53. In cases where compensation for inconveniences, damage of crops etc. are claimed by the owner, the Client has to be informed by the Contractor through the SE. This compensation is in general settled under the responsibility of the Client before signing the Contract. In unforeseeable cases, the respective administrative entities of the Client will take care of compensation.

Contractor's Environment, Health and Safety Management Plan (EHS-MP)
54. Within 6 weeks of signing the Contract, the Contractor shall prepare an EHS-MP to ensure the adequate management of the health, safety, environmental and social aspects of the works, including implementation of the requirements of these general conditions and any specific requirements of an EMP for the works. The Contractor's EHS-MP will serve two main purposes:

- For the Contractor, for internal purposes, to ensure that all measures are in place for adequate EHS management, and as an operational manual for his staff.
- For the Client, supported where necessary by a SE, to ensure that the Contractor is fully prepared for the adequate management of the EHS aspects of the project, and as a basis for monitoring of the Contractor's EHS performance.

55. The Contractor's EHS-MP shall provide at least:

- a description of procedures and methods for complying with these general environmental management conditions, and any specific conditions specified in an EMP;
- a description of specific mitigation measures that will be implemented in order to minimize adverse impacts;

- a description of all planned monitoring activities (e.g. sediment discharges from borrow areas) and the reporting thereof; and
- the internal organizational, management and reporting mechanisms put in place for such.

56. The Contractor's EHS-MP will be reviewed and approved by the Client before start of the works. This review should demonstrate if the Contractor's EHS-MP covers all of the identified impacts, and has defined appropriate measures to counteract any potential impacts.

EHS Reporting

57. The Contractor shall prepare bi-weekly progress reports to the SE on compliance with these general conditions, the project EMP if any, and his own EHS-MP. An example format for a Contractor EHS report is portrayed below. It is expected that the Contractor's reports will include information on:

- EHS management actions/measures taken, including approvals sought from local or national authorities;
- Problems encountered in relation to EHS aspects (incidents, including delays, cost consequences, etc. as a result thereof);
- Lack of compliance with contract requirements on the part of the Contractor;
- Changes of assumptions, conditions, measures, designs and actual works in relation to EHS aspects; and
- Observations, concerns raised and/or decisions taken with regard to EHS management during site meetings.

58. It is advisable that reporting of significant EHS incidents be done "as soon as practicable". Such incident reporting shall therefore be done individually. Also, it is advisable that the Contractor keep his own records on health, safety and welfare of persons, and damage to property. It is advisable to include such records, as well as copies of incident reports, as appendixes to the bi-weekly reports. A sample format for an incident notification is shown below. Details of EHS performance will be reported to the Client through the SE's reports to the Client.

Training of Contractor's Personnel

59. The Contractor shall provide sufficient training to his own personnel to ensure that they are all aware of the relevant aspects of these general conditions, any project EMP, and his own EHS-MP, and are able to fulfil their expected roles and functions. Specific training should be provided to those employees that have particular responsibilities associated with the implementation of the EHS-MP. General topics should be:

- EHS in general (working procedures);
- emergency procedures; and
- social and cultural aspects (awareness raising on social issues).

Cost of Compliance

60. It is expected that compliance with these conditions is already part of standard good workmanship and state of art as generally required under this Contract. The item "Compliance with Environmental Management Conditions" in the Bill of Quantities covers this cost. No other payments will be made to the Contractor for compliance with any request to avoid and/or mitigate an avoidable EHS impact.

3. Example Format: EHS Report

Contract: Period of reporting:

EHS management actions/measures:

Summarize EHS management actions/measures taken during period of reporting, including planning and management activities (e.g. risk and impact assessments), EHS training, specific design and work measures taken, etc.

EHS incidents:
Report on any problems encountered in relation to EHS aspects, including its consequences (delays, costs) and corrective measures taken. Include relevant incident reports.
EHS compliance:
Report on compliance with Contract EHS conditions, including any cases of non-compliance.
Changes:
Report on any changes of assumptions, conditions, measures, designs and actual works in relation to EHS aspects.
Concerns and observations:
Report on any observations, concerns raised and/or decisions taken with regard to EHS management during site meetings and visits.

Signature (Name, Title Date):
Contractor Representative

4. Example Format: EHS Incident Notification

EHS Incident Notification

Provide within 24 hrs to the Supervising Engineer
Originators Reference No:.....

Date of Incident:..... Time:.....
Location of incident:.....
Name of Person(s) involved:.....
Employing Company:.....
Type of Incident:.....
Description of Incident:
Where, when, what, how, who, operation in progress at the time (only factual)
Immediate Action:
Immediate remedial action and actions taken to prevent reoccurrence or escalation

Signature (Name, Title, Date):.....
Contractor Representative

Appendix 5 : Project Occupational Health and Safety (OHS) Plan

INTRODUCTION

Every project poses its HSE risks including the ACE-SPED project. This plan was necessitated to meet up with OHS standards and to achieve the objectives set for the proposed project. The OHS will be applicable to the entire project phases. The project team shall undertake to ensure high performance standards and conformity with contract requirements by managing the works in a systematic and thorough manner.

Key components of the OHS plan	Description
Competency	All personnel required to operate or work with any equipment or machine must be tested for each equipment he/ she must be operating. All personnel who as part of their profession require licensing or certification must obtain the necessary certification before he/she shall be allowed to work on the site.
Fitness	All personnel working on site shall be required to be certified medically fit to do so by an approved medical facility or Medical Doctor (pre-employment medical examination)
HSE Training : Induction/ Orientation	Every new or rehire employee and subcontractors employees must undergo mandatory OHS orientation/ induction. The purpose of the induction is to educate workers and make them aware of the major potential hazards he or she shall come into contact with while working on the site; also, it is one more opportunity to stress the importance of HSE being the first priority in the operation. The content of the HSE orientation/ induction shall cover the following subjects: • Site safety rules • Personnel protective equipment requirements (PPE) • Environmental sensitivity and protection • Preparation and planning of the job (Daily Pre-tasks talk) • Emergency plan and muster points.
Project specific training	In addition to the HSE orientation/induction, there shall be specific HSE trainings which shall cover the following topics • Manual handling • Electrical safety • Emergency prevention, preparedness and response • Work at height training • First Aid training (for site First Aiders) • Safe driving techniques (for drivers)
Hazard identification & HSE risk assessment	The project HSE risk assessment shall be developed and recorded. The project's HSE risk assessment shall be conducted by a team consisting of the HSE Manager/ Supervisor and technical managers/ supervisors. It must be approved by the project manager.
Fire Risk Assessment	A fire risk assessment shall be developed and recorded. A fire safety plan shall be in place in the site
Job Hazard Analysis	Job hazard analysis is required when the hazards and risks associated with a specific task is to be identified so as to implement control measures. The

	HSE department together with the technical managers/supervisors shall develop a job hazard analysis when applicable
Emergency Preparedness And Response	Emergency procedures and evacuation plan shall be developed by the HSE Department and displayed on the notice board. These procedures shall be communicated to all staff. Also, each section/department shall have at least a trained first aider at all times.
HSE IMPLEMENTATION AND PERFORMANCE MONITORING	
HSE Meetings	HSE management meetings shall be held once a month. The meeting is to help identify safety problems, develop solutions, review incident reports, provide training and evaluate the effectiveness of our safety program. Some of the meetings shall be: • Project/Site Management HSE Meeting for management and supervision (Monthly). • Tool box talk meetings for all workforce (Weekly). • Pre-task briefing for all workforces (Daily). Special situation meeting (As required).
HSE Reporting	All incidents and illnesses must be reported to site supervisor after which investigation shall commence and recorded so that appropriate corrective actions shall be implemented to prevent any re-occurrence and report findings shall be forwarded to management for review. Reporting requirements shall include notification of incident, investigation report, and monthly report. Notification of Incident form shall be developed which shall be filled and submitted to HSE department for investigation.
HSE Inspection and Audits	For continual improvement of HSE management system, HSE inspection and audit shall be conducted. An inspection checklist shall be developed. This is to ensure that the HSE management system is being adhered to. The inspection shall be conducted by the HSE department together with site management.
Corrective and Preventive Actions and Non Conformities	During the cause of inspections, concerns raised shall be addressed and closed out. It is expected that in a period of two weeks, a close out inspection shall take place to verify that the corrective actions have been closed.
Project HSE Rules	The project HSE rules shall be developed and supervision shall develop specific rules and procedures when necessary. The following site rules shall be implemented at all times. The Site Manager shall draw these rules to the attention of their own workmen or staff. All subcontractors must ensure that these rules are drawn to the attention of their workmen and staff. The Principal Contractor may implement additional site rules during the contract programme. Any such additional rules shall be notified to all personnel engaged on the project prior to their implementation. The HSE rules shall include but not limited to: 1. Personal Protective Equipment must be worn at all times. 2. All instructions issued by the Site Manager regarding the storage, handling or cleaning of materials, plant and equipment must be followed. 3. All vehicles must be parked in the designated areas.

	4. Any workman suffering from a medical condition that might affect his work and/or that could require specific Medical treatment must inform the supervisor before commencing work. 5. All site tools shall either be battery operated or 110 volts. 6. No one shall be permitted on site if it is believed that they are under the influence of alcohol or drugs. 7. Vehicles must not reverse without a banksman in attendance. 8. All visitors to site must undergo a site-specific induction and operative Identity badges must be worn at all times. 9. All excavations must be secured. 10. Smoking and eating shall only be permitted in the designated area. This area shall be identified during induction. 11. No hot works operations are permitted without a hot work permit in place. 12. There shall be no radios or other music playing devices on site. 13. Good housekeeping practices to be adopted**.** 14. Compliance with all Ethical Power Permit to Work systems 15. The site keyed access procedure must be strictly adhered to. 16. All Contractors must comply with Site Health & Safety Guidelines / Site Safety Method Statement 17. No untrained worker shall be permitted to operate heavy machineries.
SAFE WORK PRACTICES	
Personal Protective Equipment (PPE)	The basic PPE required for the project shall be Safety Glasses, nose mask, Safety Boots, Hand Gloves, Hard Hat and Coverall. Any other PPE shall be used as applicable. Management is responsible for the provision of PPE and usage shall be enforced at all time. PPE shall be provided in circumstances where exposure to hazards cannot be avoided by other means or to supplement existing control measures identified by a risk assessment. An assessment shall be made to ensure that the PPE is suitable for purpose and is appropriate to the risk involved. Information, instruction & training shall be given to all employees on safe use, maintenance and storage of PPE. Employees shall, in accordance with instructions given, make full use of all PPE provided and maintain it in a serviceable condition and report its loss or defect immediately to the maintenance department where it shall be replaced. PPE shall be replaced when it is no longer serviceable and returned on a new for old basis. Employees shall sign to state that they have received PPE when issued.
Welfare Facilities	The provision of welfare facilities on the site shall be communicated to all operatives at site induction. A cleaning regime shall be implemented and maintained for the duration of the construction phase to ensure the site welfare facilities remain in a clean and tidy condition. If mains drinking water becomes unavailable during the construction phase bottled water shall be brought to site for all operatives for the necessary period.
Signage	Adequate provision for warning and directional signs shall be made

Project HSE Procedures	OHS procedures shall be developed. Project activities shall generally be controlled in accordance with OHS Procedures. These procedures shall include: • HSE Reporting Procedure • Working at Height Procedure. • Emergency Procedure.

Key roles	**Responsibilities**
Project manager	• Set good example in HSE issues. • Ensure the availability of resources essential to establish, implement, maintain and improve the OHS Management System. • Define, document and communicate roles, allocate responsibilities and accountabilities, delegating authorities, to facilitate effective OHS management. • Ensure that all of the activities undertaken in the Project conform to Nigerian legislation, client requirements or international standards when applicable. • Review objectives achievements throughout the year
Project supervisors	• Enforcing all phases of the established HSE plan. • Set good example in HSE issues. • Preparing Job Hazard Analysis when required. • Ensuring the safety of all workers associated with the site. • Conducting HSE inspections. • Ensuring workers are competent for their allocated tasks. • Attending and participating in HSE meetings. • Participating in accident investigations.
HSE Manager/ Supervisor	• Prepare relevant OHS documentation and procedures. • Monitor the efficient implementation of OHS requirements. • Participate and organize the OHS risk assessments. • Advise management of compliance and of conditions requiring attention. • Conduct regular HSE inspections. • Make thorough analysis of statistical data and inspections; delineates problem areas; and makes recommendation for solutions. • Take part in the review of all OHS incidents and assist in investigating incident. • Monitor the efficient implementation of the Project's OHS requirements. • Organize the Project's OHS risk assessment exercises. • Check on the use of all types of personal protective equipment specifies the use of appropriate PPE for the various work activities. Evaluates their effectiveness and suggests improvements where indicate
HSE Advisor	• Check on the use of all types of personal protective equipment specifies the use of appropriate PPE for the various work activities. Evaluates their effectiveness and suggests improvements. • Conduct independent inspections to observe conformance with established OHS Plan and determines the effectiveness of individual elements of the plan (pre-task briefing, weekly toolbox talk, etc)

	• Establish contact with Subcontractors with the objective of maintaining good relations and coordination of accident prevention activities and compliance with the established OHS plan. • Correct unsafe acts and unsafe conditions. • Deliver HSE induction/orientation course to all employees, including subcontractors. • Deliver HSE awareness course and toolbox talk. • Advise employees on OHS matters.
Employees	• Take all reasonable and practical steps to care for their own health and safety and avoid affecting the health and safety of co-workers and the general public. • Follow all instructions and use the equipment properly • Not interfere with any safety arrangements. • Report any circumstances which may not comply with the project's OHS management system.

Appendix 6 : Employee Code of Conduct (CoC) Policy

This Employee Code of Conduct is meant to provide general guidelines and should be used as a reference to guide the development of Contractor in developing Code of Conduct for workers that will be involved in the Construction and Renovation works in the UNN ACE-SPED project.

Policy brief & purpose

The policy brief and purpose is to outline the expectations of the Contractor regarding all employees' behaviour towards the company, their colleagues, students, females, supervisors, sub-contractors and overall organization as well as clients, third parties and host communities. Emphasis should be given to freedom of expression and open communication and gender base violence and sexual exploitation. In addition, all employees must be bound to follow the code of conduct.

Scope

The policy applies to all employees regardless of employment agreement or rank.

Policy elements

All employees must be bound by their contract to follow the Code of Conduct while performing their duties. As a minimum, the code of conduct should include the components outline below:

- **Compliance with law**: All employees must protect the company's legality. They should comply with all environmental, safety and fair dealing laws. It is expected that employees should be ethical and responsible when dealing with company's equipment, products, clients, host communities, females and public image.
- **Respect in the workplace**: All employees should respect their colleagues. Any kind of discriminatory behavior, harassment or victimization should be prohibited. Employees should conform with our equal opportunity policy in all aspects of their work, from recruitment and performance evaluation to interpersonal relations.
- **Protection of Company Property**: All employees should treat company's property, whether material or intangible, with respect and care.

  Employees:

  (i) Shouldn't misuse company equipment or use it frivolously.

  (ii) Should respect all kinds of incorporeal property. This includes trademarks, copyright and other property (information, reports etc.) Employees should use them only to complete their job duties.

  (iii) Should protect company facilities and other material property (e.g. company cars) from damage and vandalism, whenever possible.
- **Professionalism**: All employees must show integrity and professionalism in the workplace.
- **Personal appearance**: All employees must follow our dress code and personal appearance guidelines including using appropriate PPE at all times.
- **Corruption:** the code of conduct should strongly discourage employees from accepting gifts from clients or partners. And, prohibit briberies for the benefit of any external or internal party.
- **Job duties and authority**: All employees should fulfill their job duties with integrity and respect toward customers, stakeholders and the community. Supervisors and managers must

not abuse their authority. We expect them to delegate duties to their team members taking into account their competences and workload. Likewise, we expect team members to follow team leaders' instructions and complete their duties with skill and in a timely manner. mentoring throughout the company should be encouraged.

- **Absenteeism and tardiness:** Employees should follow their schedules. We can make exceptions for occasions that prevent employees from following standard working hours or days. But, generally, we expect employees to be punctual when coming to and leaving from work.
- **Conflict of interest:** it is expected that employees should avoid any personal, financial or other interests that might hinder their capability or willingness to perform their job duties.
- **Collaboration**: Employees should be friendly and collaborative. They should try not to disrupt the workplace or present obstacles to their colleagues' work.
- **Communication:** All employees must be open for communication with their colleagues, supervisors or team members.
- **Relationship with host communities:** The code of conduct should promote a healthy relationship between workers and and the university community. Measures to facilitate such relationship include but not limited to Respect for local residents; ethical conducts,; Zero tolerance of illegal activities such as child sexual exploitation and underage sex, prostitution, harassment of women, GBV, purchase or use of illegal drugs.
- **Benefits:** We expect employees to not abuse their employment benefits. This can refer to time off, insurance, facilities, subscriptions or other benefits our company offers.
- **Policies**: All employees should read and follow the company policies. If they have any questions, they should ask their managers or Human Resources (HR) department.
- **Disciplinary actions**: The company should take disciplinary action against employees who repeatedly or intentionally fail to follow the code of conduct. Disciplinary actions will vary depending on the violation. Possible consequences include:
  - Demotion.
  - Reprimand.
  - Suspension or termination for more serious offenses.
  - Detraction of benefits for a definite or indefinite time.

The code of conduct should provide detailed disciplinary measures and sanctions (e.g. dismissal) for infringement of the code of conduct and/or company rules; Commitment / policy to cooperate with law enforcement agencies investigating perpetrators of gender-based violence.
The code of conduct for all workers must be included in employees handbook and must also be given to all workers

Appendix 7: Waste Management Plan

Activity Description	Action to be taken	Timing (When it will be done)	Responsibility (Who to do it)	Budget	Monitoring indicator
On-site waste management during construction and renovation	▪ Liaise with ESWAMA to discuss on evacuation and disposal of waste to be generated by the project activity ▪ Hold prior consultations with ESWAMA through the PIU	During mobilization phase	Contractor	Part of contractor's operational cost	Meeting report
	▪ Identify, demarcate, restrict area where waste will be collected on-site ▪ All waste categories including e-waste should be separated ▪ All categories of waste should not be dumped together ▪ Dispose all wastes including e-waste in authorized areas, metals, used oils, etc. ▪ Establish and enforce daily site clean-up procedures, including maintenance of adequate disposal facilities for construction debris	During construction	contractor	(a) 500,000	Site inspection
	Spoil and topsoil are to be stockpiled onsite and mitigation measures for dust control in the ESMP table to be implemented	During construction	contractor	To be captured in (a) above Also, part of mitigation costs	Site inspection

	Liquid waste is to be stored in appropriate containers in bunded areas until transported offsite. Bunded areas will have the capacity to over 100% of the waste volume for bulk storage.	During construction	contractor	Captured in (a) above	Site inspection
	Ensure that all equipment maintenance activities, including oil changes, are conducted within demarcated maintenance areas designated for such. Ensure that oil and other lubricants are never dumped on the ground, but only in designated areas.	During construction	contractor	Captured in (a) above	Site inspection
	▪ Construction material accumulation shall be reduced by any possibility ▪ All e-wastes must be cleared from site before final demobilization from site ▪ Good housekeeping procedure to be followed ▪ No waste must be burnt on-site	During construction/demobilization from site	contractor	Captured in (a) above	Site inspection
Waste Evacuation	▪ Prior 7 days notification to ESWAMA of waste to be generated through PIU	During construction	Contractor	Part of operational cost	notice

and transportation	▪ Transport vehicles must meet regulatory specifications ▪ Loads must be handled, unloaded and placed in the cell carefully ▪ Use of appropriate PPEs (disposable nose mask, gloves, coverall, boot, eye goggle etc) ▪ Drivers/personnel must be trained personnel on waste management ▪ A weighbridge attendant must be present at evacuation of waste from site ▪ Wastes should be transported to a licensed waste management facility	During construction	ESWAMA	Captured under capacity building cost in ESMP implementation budget	Visual observation and inspection
Waste disposal	▪ Relocate waste including e-waste to disposal site ▪ Dispose waste including e-waste in an environmentally acceptable manner in line with national and State waste management Enugu State waste management regulations	During construction	contractor	Captured in (a) above	Site inspection
Manage waste in project operation phase	▪ Provide waste bins in the ACE-SPED Centre for each category of waste including e-waste ▪ Provide waste bins in the hostel for each category of waste in the student hostel ▪ Liaise with ESWAMA on periodic waste evacuation from the Centre and the hostels ▪ Ensure good housekeeping practices	During operation phase	ACE-SPED management	Part of the yearly operational cost for ACE-SPED	Visual observation

Appendix 8: Traffic Management Plan (TMP)

The objective of this TMP is to provide safe passage for pedestrians, motorcyclist, cyclists and vehicular traffic along the University project corridor during the construction and rehabilitation civil works.
Cost of the TMP is embedded into the contract cost

S/N	Aspects	Descriptions	Responsible Party	Cost
1	Traffic/Safety Signage	▪ Safety signage should be put at strategic locations from the location of the borrow pit in town to the site in UNNroad to warn road users of the ongoing construction activities. ▪ Signages should also be located along borrow pits, engineering yards and workers' camp.	Contractor	100,000
2		▪ Mobilization of equipment and materials should be done at off-peak period (10am – 4pm). ▪ Enforce speed limit. ▪ Ensure vehicles and equipment are parked at Camp site and designated areas ONLY. ▪ Untarred access roads shall be sprinkled with water at least 2 times a day to suppress dust emissions. ▪ The contractor must ensure that trucks carrying sand/soil to and from the sites are well covered in order not to cause injury to the public.	Contractor	-- 100,000
3	Training	▪ Hire drivers with appropriate driver's license. ▪ As part of refresher course for construction workers, train drivers on defensive driving and enforce speed limits	Contractor	200,000
4	Communication	▪ All Traffic and Safety signages should be boldly written in English & local languages. ▪ The Contractor, with support from the PIU should undertake road safety campaigns in communities along the road to sensitize them about the project activities. ▪ Any incident/ accidents should be reported immediately to the PIU	Contractor PIU Contractor	See 1 Above 100,000
	Cost	All cost included have been embedded in the ESMP Matrix Table 6.9		***500,000***

Appendix 9: Chance Finds Procedure

Purpose of the chance find procedure

The chance find procedure is a project-specific procedure that outlines actions required if previously unknown heritage resources, particularly archaeological resources, are encountered during project construction or operation. A Chance Find Procedure, as described in IFC Performance Standard 8 and EBRD Performance Requirement 8 and law on Cultural Heritage, is a process that prevents chance finds from being disturbed until an assessment by a competent specialist is made and actions consistent with the requirements are implemented.

Scope of the chance find procedure

This procedure is applicable to all activities conducted by the personnel, including contractors, that have the potential to uncover a heritage item/site. The procedure details the actions to be taken when a previously unidentified and potential heritage item/site is found during construction activities. Procedure outlines the roles and responsibilities and the response times required from both project staff, and any relevant heritage authority.

Induction/Training

All personnel, especially those working on earth movements and excavations, are to be inducted on the identification of potential heritage items/sites and the relevant actions for them with regards to this procedure during the Project induction and regular toolbox talks.

Chance find procedure

If any person discovers a physical cultural resource, such as (but not limited to) archaeological sites, historical sites, remains and objects, or a cemetery and/or individual graves during excavation or construction, the following steps shall be taken:

1. Stop all works in the vicinity of the find, until a solution is found for the preservation of these artefacts, or advice from the relevant authorities is obtained;
2. Immediately notify a foreman. The foreman will then notify the Construction Manager and the Environment Officer (EO)/Environmental Manager (EM);
3. Record details in Incident Report and take photos of the find;
4. Delineate the discovered site or area; secure the site to prevent any damage or loss of removable objects. In cases of removable antiquities or sensitive remains, a night guard shall be arranged until the responsible local authorities take over;
5. Preliminary evaluation of the findings by archaeologists. The archaeologist must make a rapid assessment of the site or find to determine its importance. Based on this assessment the appropriate strategy can be implemented. The significance and importance of the findings should be assessed according to the various criteria relevant to cultural heritage such as aesthetic, historic, scientific or research, social and economic values of the find;
6. Sites of minor significance (such as isolated or unclear features, and isolated finds) should be recorded immediately by the archaeologist, thus causing a minimum disruption to the work schedule of the Contractor. The results of all archaeological work must be reported to the Ministry of Culture/Agency in charge of National Heritage, once completed.
7. In case of significant find, the Agency/Ministry (Agency for Protection of National Heritage or Archaeological Research Centre, hereinafter referred to as Heritage team) should be

informed immediately and in writing within 7 days from the find (ref.law on heritage protection).

8. The onsite archaeologist provides the Heritage team with photos, other information as relevant for identification and assessment of the significance of heritage items.
9. The Ministry must investigate the fact within 2 weeks from the date of notification and provide response in writing.
10. Decisions on how to handle the finding shall be taken by the responsible authorities. This could include changes in the layout (such as when finding an irremovable remain of cultural or archaeological importance) conservation, preservation, restoration and salvage;
11. Construction works could resume only after permission is granted from the responsible authorities.
12. In case no response received within the 2 weeks period mentioned above, this is considered as authorisation to proceed with suspended construction works.

One of the main requirements of the procedure is record keeping. All finds must be registered. Photolog, copies of communication with decision making authorities, conclusions and recommendations/guidance, implementation reports – kept.

Additional information Management options for archaeological site

- **Site avoidance.** If the boundaries of the site have been delineated attempt must be made to redesign the proposed development to avoid the site. (The fastest and most cost-effective management option)
- **Mitigation.** If it is not feasible to avoid the site through redesign, it will be necessary to sample it using data collection program prior to its loss. This could include surface collection and/or excavation. (The most expensive and time-consuming management option.)
- **Site Protection.** It may be possible to protect the site through the installation of barriers during the time of the development and/or possibly for a longer term. This could include the erection of high visibility fencing around the site or covering the site area with a geotextile and then capping it with fill. The exact prescription would be site- specific.

Management of replicable and non-replicable heritage

Different approaches for the finds apply to replicable and non-replicable heritage.

Replicable heritage

Where tangible cultural heritage that is replicable[1] and not critical is encountered, mitigation measures will be applied.

The mitigation hierarchy is as follows:

- Avoidance;
- Minimization of adverse impacts and implementation of restoration measures, in situ;

[1] Replicable cultural heritage is defined as tangible forms of cultural heritage that can themselves be moved to another location or that can be replaced by a similar structure or natural features to which the cultural values can be transferred by appropriate measures. Archaeological or historical sites may be considered replicable where the particular eras and cultural values they represent are well represented by other sites and/or structures.

- Restoration of the functionality of the cultural heritage, in a different location;
- Permanent removal of historical and archaeological artefacts and structures ;
- Compensation of loss - where minimization of adverse impacts and restoration not feasible.

Non-replicable heritage

Most cultural heritage is best protected by in situ preservation, since removal is likely to result in irreparable damage or even destruction of the cultural heritage.

Non-replicable cultural heritage[2] must not be removed unless all of the following conditions are met:

- There are no technically or financially feasible alternatives to removal;
- The overall benefits of the project conclusively outweigh the anticipated cultural heritage loss from removal; and

Any removal of cultural heritage must be conducted using the best available technique advised by relevant authority and supervised by archaeologist.

Human Remains Management Options

The handling of human remains believed to be archaeological in nature requires communication according to the same procedure described above.

There are two possible courses of action:

- **Avoid.** The development project is redesigned to completely avoid the found remains. An assessment should be made as to whether the remains may be affected by residual or accumulative impacts associated with the development, and properly addressed by a comprehensive management plan.
- **Exhumate.** Exhumation of the remains in a manner considered appropriate by decision makers. This will involve the predetermination of a site suitable for the reburial of the remains. Certain ceremonies or procedures may need to be followed before development activities can recommence in the area of the discovery.

[2] Nonreplicable cultural heritage may relate to the social, economic, cultural, environmental, and climatic conditions of past peoples, their evolving ecologies, adaptive strategies, and early forms of environmental management, where the (i) cultural heritage is unique or relatively unique for the period it represents, or (ii) cultural heritage is unique or relatively unique in linking several periods in the same site. Examples of non-replicable cultural heritage may include an ancient city or temple, or a site unique in the period that it represents.